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M.Tech AI

Game Theory

April 12, 2025

1 Introduction

- Test 1 on Feb 19(20), Test 2 on Apr 2(20)
- Miniproject Finalization by Jan 31, Midterm evaluation on Feb 15(5), Submission on Apr 6(10) and Final Evaluation on Apr 11(5)
- Final Exam of 30 marks
- Game Theory and Mechanism Design by Y. Narahari, 2014
- A rich source of material on game theory and game theory courses, `http://www.gametheory.net`.
- Course material and several survey articles can be downloaded, `http://gtl.csa.iisc.ac.in/gametheory/`.

2 Non-Cooperative Games

2.1 Strategic Form Game/Normal Form Game

- Consists of a set of N players, $N = \{1, 2, ..., n\}$
- **Strategy set/Action set/Pure Strategy set** $(S_i)_{i \in N}$
- Set of all strategy profiles/vectors: $S = S_1 \times S_2 \times ... \times S_n$
- **Utility/Reward/Payoff**: $u_i : S \to \mathbb{R}$
- Game is defined by the triple $\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$
- This is a **Simultaneous move game**, all players will move simultaneously.
- **Non-Cooperative Game**: Players are not aware of each other's move.
- **Rationality**: Expected utility maximization, each player strives to maximize their own utility.
- **Intelligent**: Every player is equally competent/knowledgeable.
- **Fundamental Assumption**: Each player is Rational and Intelligent, Each player knows the game, strategies, utility function, Each player knows that each player knows the game, Each player knows that each player knows that each player knows the game, ad infinitum.
- This common knowledge is also included in Intelligence.
- **Matching Pennies Game**: There are 2 players, each have a coin. They both toss a coin independently. Pure Strategy Nash Equilibrium does not exist for this game.
- **Prisoner's Dilemma**: There are 2 players caught allegedly committing crime. They are being interrogated separately. They can either confess(C) or not confess(NC). (C, C) is a Strongly Dominated Strategy Equilibrium, and a PSNE.

1 \ 2	H	T
H	$1, -1$	$-1, 1$
T	$-1, 1$	$1, -1$

(a) Matching Pennies

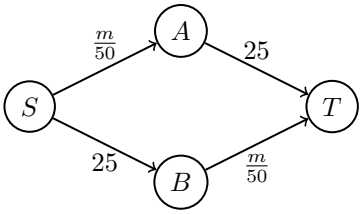
1 \ 2	NC	C
NC	$-2, -2$	$-10, -1$
C	$-1, -10$	$-5, -5$

(b) Prisoner's Dilemma

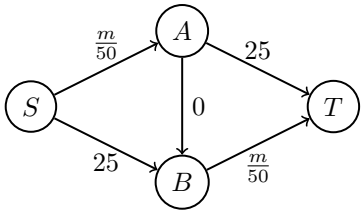
Figure 1: Normal Form Game Examples

- **Finite Games**: Number of players are finite, and strategy set is also finite.
- **Infinite Games**: Number of players can be infinite or at least 1 strategy set is infinite.

- **Braess Paradox (Traffic)**: There is a source S and a destination T, there are also 2 hubs A, and B. Consider the following two different games



(a) Without a Bridge



(b) With a Bridge

Consider 1000 vehicles that need to choose between a path, $N = \{1, 2, ..., 1000\}$
In the first case, each vehicle will have the strategy set $S_1 = S_2 = ... = S_{1000} = \{A, B\}$
$n_A(s_1, ..., s_n)$ = number of vehicles that chose A, similarly define n_B for B, $n_A + n_B = n$

$$u_i(s_1, ..., s_n) = \begin{cases} -(25 + \frac{n_A}{50}), \text{ if } s_i = A \\ -(25 + \frac{n_B}{50}), \text{ if } s_i = B \end{cases}$$

In this case, 500 choosing A and 500 choosing B happens to be pure strategy Nash equilibrium.
In the second case, each vehicle will have the strategy set $S_1 = S_2 = ... = S_{1000} = \{A, B, AB\}$
Now we will define n_A, n_B and n_{AB} same as before.

$$u_i(s_1, ..., s_n) = \begin{cases} -(25 + \frac{(n_A+n_{AB})}{50}), \text{ if } s_i = A \\ -(25 + \frac{(n_B+n_{AB})}{50}), \text{ if } s_i = B \\ -(\frac{(n_A+n_{AB})}{50} + 0 + \frac{(n_B+n_{AB})}{50}), \text{ if } s_i = AB \end{cases}$$

All vehicles choosing AB happens to be a strongly dominant strategy equilibrium.
Adding an additional pathway leads to increased congestion!

- **Auction**: A seller wants to sell a single, indivisible object to a prospective buyer. There are n buyers, each of which have a certain valuation v_i they have for the object and each bid s_i. $S_i \in \mathbb{R}^+$
$(s_1, ..., s_n)$ is the bid profile $\in S_1 \times \cdots \times S_n$
Highest bid is the winner, in case of tie the lower index wins.
Player i only knows its own valuation v_i, all other valuations are private information.
We define the following two variables

$$y_i(s_1, ..., s_n) = 1 \text{ if } i \text{ wins else } 0 = \begin{cases} 1, \text{ if } s_i > s_j \forall j \in \{1, ..., i-1\} \wedge s_i \geq s_j \forall j \in \{i+1, ..., n\} \\ 0, \text{ otherwise} \end{cases}$$

$$t_i(s_1, ..., s_n) = \text{The payment that } i \text{ makes if they win}$$

Finally, we can define the utility function based on the valuation as

$$u_i(s_1, ..., s_n) = y_i(s_1, ..., s_n)(v_i - t_i(s_1, ..., s_n))$$

This is called **single item single bid auction**.

2.2 Pure Strategies

- We define a vector $S_{-i} = S_1 \times ... \times S_{i-1} \times S_{i+1} \times ... \times S_n = \times_{j \in N, j \neq i} S_j$
- $s_{-i} \in S_{-i}$, $s_{-i} = (s_1, ..., s_{i-1}, s_{i+1}, ..., s_n)$
- Complete strategy $s \in S$, $s = (s_1, s_2, ..., s_n) = (s_i, s_{-i})$
- **Strongly Dominant Strategy(SDS)**: $s'_i \in S_i$ is said to strongly dominate another strategy $s'_i \in S_i$ if

$$u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i}), \ \forall s_{-i} \in S_{-i}$$

- If s'_i strongly dominates $\forall s_i \in S_i$, $s_i \neq s'_i$ then s'_i is SDS.
- We say s'_i **very weakly dominates** s_i if $u_i(s'_i, s_{-i}) \geq u_i(s_i, s_{-i}), \ \forall s_{-i} \in S_{-i}$
- If s'_i very weakly dominates $\forall s_i \in S_i$, $s_i \neq s'_i$ then s'_i is a **very weakly dominant strategy(VWDS)**.
- s'_i **weakly dominates** s_i if s'_i very weakly dominates s_i and $\exists s_{-i} \in S_{-i}$ such that

$$u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i})$$

- If $\exists$ a profile $(s_1^*, s_2^*, ..., s_n^*)$ such that s_i^* is a SDS(WDS)(VWDS) for player i, $\forall i \in N$, then such a profile is called SDSE(WDSE)(VWDSE)

- **Price of Anarchy**: SDS profile may not maximize social welfare.

- Consider the second part of Braess Paradox, with a bridge 2b
  We will show that $(AB, ...AB)$ is a SDSE
  Suppose $s_{-i} \in S_{-i}$ and $n_A(s_{-i}), n_B(s_{-i}), n_{AB}(s_{-i})$ be appropriate numbers

$$\begin{aligned}
u_i(AB, s_{-i}) &= -\left(\frac{n_A(s_{-i}) + n_{AB}(s_{-i}) + 1}{50} + 0 + \frac{n_B(s_{-i}) + n_{AB}(s_{-i}) + 1}{50}\right) \\
&= -\left(\frac{n_A(s_{-i}) + n_{AB}(s_{-i}) + n_B(s_{-i}) + 1}{50} + \frac{n_{AB}(s_{-i}) + 1}{50}\right) \\
&= -(\qquad 20 \qquad + \frac{n_{AB}(s_{-i}) + 1}{50}) \\
u_i(A, s_{-i}) &= -\left(\frac{n_A(s_{-i}) + n_{AB}(s_{-i}) + 1}{50} + 25\right) \\
u_i(B, s_{-i}) &= -\left(25 + \frac{n_B(s_{-i}) + n_{AB}(s_{-i}) + 1}{50}\right)
\end{aligned}$$

$$\implies u_i(AB, s_{-i}) > u_i(A, s_{-i}) \text{ and } u_i(AB, s_{-i}) > u_i(B, s_{-i}) \ \forall s_{-i} \in S_{-i}$$

  Hence, $(AB, AB, ..., AB)$ is a SDSE.

- **Vickry Auction**: Also known as **Second Price Auction**, the winner pays the second-highest bid or the highest loosing bid. Consider the previous definition of auction and let the bids be $(b_1, b_2, ..., b_n)$.
  We will show that $(b_1 = v_1, b_2 = v_2, ..., b_n = v_n)$ is a WDSE.
  For each $i \in N$, we want to show $b_i = v_i$ is a WDS
  Essentially, $\forall b_i'(\neq v_i)$ we want to show

$$u_i(v_i, b_{-i}) \geq u_i(b_i', b_{-i}) \ \forall b_{-i} \in S_{-i} \text{ and}$$
$$\exists \text{ some } b_{-i} \in S_{-i} \text{ such that } u_i(v_i, b_{-i}) > u_i(b_i', b_{-i})$$

Without Loss of Generality(WLoG) we will show for player 1, $b_{-1} = (b_2, b_3, ..., b_n)$

$v_1 \geq \max(b_2, ..., b_n)$ $\qquad$ $v_1 < \max(b_2, ..., b_n)$

$\underbrace{b_1 \geq \max(b_2, ..., b_n)}_{\substack{\text{1 wins} \\ u_1 = 1(v_1 - \max(b_2, ..., b_n))}}$ $\quad$ $\underbrace{b_1 < \max(b_2, ..., b_n)}_{\substack{\text{1 looses} \\ u_1 = 0}}$ $\quad$ $\underbrace{b_1 \geq \max(b_2, ..., b_n)}_{\substack{\text{1 wins} \\ u_1 = 1(v_1 - \max(b_2, ..., b_n))}}$ $\quad$ $\underbrace{b_1 < \max(b_2, ..., b_n)}_{\substack{\text{1 looses} \\ u_1 = 0}}$

$u_1 \geq 0$ $\qquad$ $u_1 \leq 0$

$\max u_1$ possible $= v_1 - \max(b_2, ..., b_n)$ $\qquad$ $\max u_1$ possible $= 0$

If player 1 bids v_1 then they will get the $\max u_1$ in both cases

Hence, proved

- **Pure Strategy Nash Equilibrium**: Given a simultaneous move game $\langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$, we say $(s_1^*, s_2^*, ..., s_n^*)$ is a PSNE iff

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \ \forall s_i \in S_i \ \forall i \in N$$

  This can also be written as $u_i(s_i^*, s_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, s_{-i}^*)$

- **Best Response Function**: Given $i \in N$ and $s_{-i} \in S_{-i}$, define $b_i(s_{-i})$ as the set of all best possible strategy when others are playing s_{-i}

$$b_i(s_{-i}) = \{s_i^* \in S_i | u_i(s_i^*, s_{-i}) \geq u_i(s_i, s_{-i}) \ \forall s_i \in S_i\}$$

- $(s_1^*, s_2^*, ..., s_n^*)$ is PSNE iff $s_i^* \in b_i(s_{-i}^*) \ \forall i \in N$

- SDSE, WDSE, and VWDSE are "clearly" PSNE

- Recall the Braess paradox without the AB junction, 2a
  Claim: $n_A = n_B = 500$ is the only PSNE.
  Consider a PSNE profile, that has n_A players choosing A and n_B players choosing B.
  Consider a player i that chooses A in the PSNE, we have to show that $u_i(A, s_{-i}) \geq u_i(B, s_{-i})$

$$-\left(25 + \frac{n_A}{50}\right) \geq -(25 + \frac{n_B + 1}{50}\Big)$$
$$n_A \leq n_B + 1$$

  Similarly, we can show this for players who choose B

$$n_B \leq n_A + 1$$

  Combining both the equations, we get

$$|n_A - n_B| \leq 1$$

  This equation is only satisfied when $n_A = n_B = 500$, as increasing n_A by 1 will make n_B decrease by 1 ending up with a difference of 2. Hence, this is the only PSNE.

- Recall the Braess paradox with the AB junction, 2b
  Claim: Any profile with $n_A, n_B > 0$ cannot be PSNE
  Consider a PSNE profile, with n_A, n_B, n_{AB} defined appropriately
  Consider a player i that chooses A in the PSNE, we have to show that $u_i(A, s_{-i}) < u_i(AB, s_{-i})$

$$-\left(\frac{n_A(s_{-i}) + n_{AB}(s_{-i}) + 1}{50} + 25\right) < -\left(\frac{n_A(s_{-i}) + n_{AB}(s_{-i}) + 1}{50} + \frac{n_B(s_{-i}) + n_{AB}(s_{-i}) + 1}{50}\right)$$
$$25 > \left(\frac{n_B(s_{-i}) + n_{AB}(s_{-i}) + 1}{50}\right) = \left(\frac{1000 - n_A(s_{-i})}{50}\right)$$
$$25 > 20 - \left(\frac{n_A(s_{-i})}{50}\right)$$

  This is obviously true, and hence the player can switch to AB to get a better utility.
  Similarly, we can show the same for a player choosing B.
  Hence, any profile with $n_A, n_B > 0$ cannot be PSNE.

- Thomas Schelling's focal point theory, deals with what will players do when there are multiple Nash equilibrium. Paper can be found at `https://www.jstor.org/stable/pdf/2235016.pdf`

- Myerson's 50th year of Nash equilibrium describes the ways in which NE was used, the paper can be found at `https://home.uchicago.edu/rmyerson/research/jelnash.pdf`.

  1. PSNE is robust only to unilateral deviation.
  2. PSNE is a good "prescription" given by a trusted mediator/referee/recommender.
  3. PSNE is a good "prediction" for what is likely to happen
  4. PSNE is like a "self-enforcing agreement" among the players
  5. PSNE is the "steady state" of an evolutionary game

- We never play pure strategies, we blend them by some probability distribution, resulting in mixed strategies.

- In finite games, it is possible that a PSNE does not exist, however at least one mixed strategy NE will always exist.

2.3 Mixed Strategies

- **Mixed Strategy**: Given a strategic form game $\langle N, (S_i), (u_i)\rangle$. Mixed strategy is defined as $\sigma_i : S_i \to [0, 1]$ such that $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$

- $\Delta(S_i)$ = Set of probability distribution on S_i, an uncountably infinite set.

- Now we define, $\langle N, (S_i), (u_i)\rangle \implies \langle N, (\Delta(S_i)), (U_i)$ where $U_i : \Delta(S_1) \times \cdots \times \Delta(S_n) \to \mathbb{R}$

- We define U_i as the expectation of joint probability distribution of all players.

- **Independent Randomization**: Players are choosing action independently of all others, hence we can make th following assumption

$$\sigma(s_1, ..., s_n) = \sigma_1(s_1)\sigma_2(s_2)\cdots\sigma_n(s_n)$$

$$U_i(\sigma_1, ..., \sigma_n) = \sum_{(s_1,...,s_n)\in S} \sigma_1(s_1)...\sigma_n(s_n)u_i(s_1, ..., s_n)$$

- We will write u instead of U, abuse of notation
- **Mixed Strategy Nash Equilibrium**: Given a strategic form game $\langle N, (\Delta(S_i)), (U_i)\rangle$, $\sigma^* = (\sigma_1^*, ..., \sigma_n^*)$ is called MSNE iff $\forall i \in N$

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*)\ \forall \sigma_i \in \Delta(S_i)$$

  This can also be written as $u_i(\sigma_i^*, \sigma_{-i}^*) = \max_{\sigma_i=\Delta(S_i)} u_i(\sigma_i, \sigma_{-i}^*)$
- **Best Response for MSNE**: Given $i \in N$ and $s_{-i} \in S_{-i}$, define $b_i(\sigma_{-i})$ as the set of all best possible strategy when others are playing σ_{-i}

$$b_i(\sigma_{-i}) = \{\sigma_i^* \in \Delta(S_i) | u_i(\sigma_i^*, \sigma_{-i}) \geq u_i(\sigma_i, \sigma_{-i})\ \forall \sigma_i \in \Delta(S_i)\}$$

- $(\sigma_1^*, ..., \sigma_n^*)$ is MSNE iff $\sigma_i^* \in b_i(\sigma_{-i}^*)\ \forall i \in N$
- All PSNE will be degenerate instances of MSNE
- **Batch & Queue Game**

1 \ 2	A	B
A	$2, 1$	$0, 0$
B	$0, 0$	$1, 2$

(a) Payoff Matrix

Clearly (A, A) and (B, B) are both PSNE
Mixed Strategy for Player 1 is defined as
$\sigma_1 = (\sigma_1(A), \sigma_1(B))$
and for Player 2 is $\sigma_2 = (\sigma_2(A), \sigma_2(B))$ such that
$\sum \sigma_1 = \sum \sigma_2 = 1$

Figure 3: Batch & Queue Game

$\Delta(S_1) = \{(\sigma_1(A), \sigma_1(B)) : \sigma_1(A), \sigma_1(B) \geq 0, \sigma_1(A) + \sigma_1(B) = 1\}$
$U_1(\sigma_1, \sigma_2) = \sigma(A, A)u_1(A, A) + \sigma(A, B)u_1(A, B) + \sigma(B, A)u_1(B, A) + \sigma(B, B)u_1(B, B)$ where $\sigma(A, A)$ is the joint probability of strategy A, A.
$u_1(\sigma_1, \sigma_2) = 2\sigma_1(A)\sigma_2(A) + \sigma_1(B)\sigma_2(B)$
$u_2(\sigma_1, \sigma_2) = \sigma_1(A)\sigma_2(A) + 2\sigma_1(B)\sigma_2(B)$
Using $\sigma_1(A) + \sigma_1(B) = 1$ and $\sigma_2(A) + \sigma_2(B) = 1$
$u_1(\sigma_1, \sigma_2) = 1 + 3\sigma_1(A)\sigma_2(A) - \sigma_1(A) - \sigma_2(A)$
$u_2(\sigma_1, \sigma_2) = 2 + 3\sigma_1(A)\sigma_2(A) - 2\sigma_1(A) - 2\sigma_2(A)$
Use the above inequalities to get the MSNE profiles $((1, 0), (1, 0)), ((0, 1), (0, 1)), ((\frac{2}{3}, \frac{1}{3}), (\frac{1}{3}, \frac{2}{3}))$

- **Convex Combination**: Consider $y_1, ..., y_n \in \mathbb{R}$, then a convex combination of this is $\lambda_1 y_1 + ... + \lambda_n y_n$ where $(\lambda_1, ..., \lambda_n)$ is a probability distribution.

$$\begin{aligned}
u_i(\sigma_i, \sigma_{-i}) &= \sum_{(s_1,...,s_n)\in S} \left(\prod_{j\in N} \sigma_j(s_j)\right) = \sum_{s_1\in S_1} \cdots \sum_{s_n\in S_n} \left(\prod_{j\in N} \sigma_j(s_j)\right) u_i(s_i, s_{-i}) \\
&= \sum_{s_i\in S_i} \sum_{s_{-i}\in S_{-i}} \left(\prod_{j\in N\setminus i} \sigma_j(s_j)\right) \sigma_i(s_i)u_i(s_i, s_{-i}) \\
&= \sum_{s_i\in S_i} \sigma_i(s_i) \sum_{s_{-i}\in S_{-i}} \left(\prod_{j\in N\setminus i} \sigma_j(s_j)\right) u_i(s_i, s_{-i}) \\
&= \sum_{s_i\in S_i} \sigma_i(s_i)u_i(s_i, \sigma_{-i})
\end{aligned}$$

- **Lemma**: Given $\Pi = \langle N, (S_i)_i, (u_i)_i\rangle$ then for any profile $(\sigma_1, \sigma_2, ..., \sigma_n)$ and for any $i \in N$

$$\max_{\sigma_i\in\Delta(S_i)} u_i(\sigma_i, \sigma_{-i}) = \max_{s_i\in S_i} u_i(s_i, \sigma_{-i})$$

  Furthermore, $P_i \in \arg\max_{\sigma_i\in\Delta(S_i)} u_i(\sigma_i, \sigma_{-i})$ iff $P_i(y) = 0\ \forall y \notin \arg\max_{s_i\in S_i} u_i(s_i, \sigma_{-i})$

- **Support**: Support of a mixed strategy σ_i is defined as

$$\delta(\sigma_i) = \{s_i \in S_i : \sigma_i(s_i) > 0\}$$

- Necessary and Sufficient condition for a mixed strategy profile to be MSNE.
  Given $\Pi = \langle N, (S_i)_{i\in N}, (u_i)_{i\in N}\rangle$, $(\sigma_1^*, \sigma_2^*, ..., \sigma_n^*)$ is a MSNE iff $\forall i \in N$

  1. $u_i(s_i\sigma_{-i}^*)$ is the same $\forall s_i \in \delta(\sigma_i^*)$
  2. $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$ $\forall s_i \in \delta(\sigma_i^*)$ $\forall s_i' \in S_i \backslash \delta(\sigma_i^*)$

- Proof of Necessity: Given $(\sigma_i^*, \sigma_{-i}^*)$ is MSNE

$$\begin{aligned}
u_i(\sigma_i^*, \sigma_{-i}^*) &= \max_{\sigma_i \in \Delta(s_i)} u_i(\sigma_i, \sigma_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*) \\
\implies \sum_{s_i \in S_i} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*) &= \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*) \\
\implies \sum_{s_i \in \delta(\sigma_i^*)} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*) &= \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*)
\end{aligned}$$

  This is similar to $\pi_1 x_1 + ... + \pi_k x_k = \max(x_1, ..., x_k, ..., x_n)$ with $\pi_1 + ... + \pi_k = 1$ and $\pi_j \neq 0$ for $j = 1, ..., k$
  This implies $x_1 = x_2 = ... = x_k = \max(x_1, x_2, ..., x_k)$
  which means $u_i(s_i, \sigma_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, \sigma_{-i}^*)$ $\forall s_i \in \delta(\sigma_i^*)$
  Clearly, this also means that

$$u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*) \ \forall s_i \in \delta(\sigma_i)^* \ \forall s_i' \in S_i \backslash \delta(\sigma_i^*)$$

- Proof of Sufficiency: Given the 2 conditions, we have to show $(\sigma_i^*, \sigma_{-i}^*)$ is a MSNE

$$u_i(\sigma_i^*, \sigma_{-i}^*) = \sum_{s_i \in S_i} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*) = \sum_{s_i \in \delta(\sigma_i^*)} \sigma_i^*(s_i) u_i(s_i, \sigma_{-i}^*)$$

  From first condition, we can write

$$\begin{aligned}
&= \sum_{s_i \in \delta(\sigma_i^*)} \sigma_i^*(s_i)\omega, \text{ where } \omega = u_i(s_i, \sigma_{-i}^*) \text{ for any } s_i \in \delta(\sigma_i^*) \\
&= \omega, \text{ since } \sum_{s_i \in \delta(\sigma_i^*)} \sigma_i^*(s_i) = 1
\end{aligned}$$

$$\sum_{s_i \in S_i} \sigma_i(s_i)\omega \geq \sum_{s_i \in S_i} \sigma_i(s_i) u_i(s_i, \sigma_{-i}^*) \text{ for any mixed strategy } \sigma_i$$

  Since, $\omega = u_i(s_i, \sigma_{-i}^*)$ for any $s_i \in \delta(\sigma_i^*)$ and is maximum by second condition, $= u_i(\sigma_i^*, \sigma_{-i}^*)$.
  Thus, $u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}^*)$ $\forall \sigma_i \in \Delta(s_i)$ which means $(\sigma_i^*, \sigma_{-i}^*)$ is a MSNE.

- **Computation of MSNE**
  Suppose $(\sigma_1, ..., \sigma_n)$ is being exposed as a possible MSNE
  $\delta(\sigma_1, ..., \sigma_n) = \delta(\sigma_1) \times \cdots \times \delta(\sigma_n)$
  Suppose we wish to explore $X_1 \times X_2 \times \cdots \times X_n$ where $X_1 \subseteq S_1$, ..., $X_n \subseteq S_n$.
  Suppose $k_i = |S_i|$ $\forall i \in N$, then the number of supports $= (2^{k_1} - 1)(2^{k_2} - 1)...(2^{k_n} - 1)$
  If $(\sigma_1, ..., \sigma_n)$ is a MSNE with the above support, then there must exist real number $\omega_1, ..., \omega_n$ such that

$$\begin{aligned}
\omega_i &= u_i(s_i, \sigma_{-i}) \ \forall s_i \in X_i \ \forall i \in N \\
&= \sum_{s_{-i} \in S_{-i}} \left(\prod_{j \in N \backslash i} \sigma_j(s_j) \right) u_i(s_i, s_{-i}) \ \forall s_i \in \delta(\sigma_i) \ \forall i \in N
\end{aligned}$$

  We need to find $\omega_1, .., \omega_n$, $\sigma_i(s_i)$ $\forall s_i \in S_i$, $\forall i \in N$
  Number of variables $= n + \sum_{i=1}^{n} k_i = n + |S_1| + |S_2| + ... + |S_n|$

Number of equations

$$\text{Support} = X_1 \times X_2 \times \cdots \times X_n = \sum_{i=1}^{N} |X_i|$$

$$\omega_i \geq u_i(s_i', \sigma_{-i}) \ \forall s_i' \in S_i \backslash X_i \ \forall i \in N = \sum_{i=1}^{n} |S_i \backslash X_i|$$

$$\sigma_i(s_i) > 0 \ \forall s_i \in X_i, \ \forall i \in N = \sum_{i=1}^{n} |X_i|$$

$$\sigma_i(s_i) = 0 \ \forall s_i \in S_i \backslash X_i, \ i \in N = \sum_{i=1}^{n} |S_i \backslash X_i|$$

$$\sum_{s_i \in X_i} \sigma_i(s_i) = 1 \ \forall i \in N = n$$

Total number of equations $= n + 2\sum_{i=1}^{n} k_i = n + 2(|S_1| + |S_2| + ... + |S_n|)$

- Lemke Howsen algorithm is used to compute MSNE in 2-player game.
- Zero-sum games can be solved using linear programming.
- Computing MSNE belongs to a class of PPAD-complete problem.
- John Nash showed that every finite strategic form game has at least one MSNE, showed using Kakutani Fixed point Theorem.
- For 2-player games, this result was shown using Browser's Fixed Point Theorem.
- More recently, this has been proved using Sperner's Lemma
- For derivation, refer to Chapter 10 of GTMD
- **Maxmin Value**: All players are trying to hurt the player i, this is the minimum guaranteed payoff the player i gets no matter what the other players do. Essentially a lower bound on utility.

$$\text{"maxmin"} \ \underline{v_i} = \max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

Suppose $(s_1^*, ..., s_n^*)$ is a PSNE

$$u_i(s_i^*, s_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, s_{-i}^*) \geq \max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) = \underline{v_i}$$

- **Minmax Value**: Essentially an upper bound on utility.

$$\text{"minmax"} \ \overline{v_i} = \min_{s_{-i} \in S_{-i}} \max_{s_i \in S_i} u_i(s_i, s_{-i})$$

Suppose s_{-i}^{minmax} is a minmax strategy against player i

$$\overline{v_i} = \max_{s_i \in S_i} u_i(s_i, s_{-i}^{minmax}) \geq \max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) = \underline{v_i}$$

Suppose $(s_1^*, ..., s_n^*)$ is a PSNE

$$u_i(s_i^*, s_{-i}^*) = \max_{s_i \in S_i} u_i(s_i, s_{-i}^*) \geq \min_{s_{-i} \in S_{-i}} \max_{s_i \in S_i} u_i(s_i, s_{-i}) = \overline{v_i}$$

$$\implies u_i(s_i^*, s_{-i}^*) \geq \overline{v_i} \geq \underline{v_i}$$

- For two player zero-sum game

$$u_i(s_i^*, s_{-i}^*) = \overline{v_i} = \underline{v_i}$$

2.4 Two Player Zero Sum Games

- Two Player Zero Sum Games(TPZS) are also known as strictly competitive games.
- TPZS games are defined by $\langle \{1,2\}, S_1, S_2, u_1, u_2 \rangle$ such that $u_1(s) + u_2(s) = 0 \ \forall s \in S_1 \times S_2$
- **Constant Sum Game**: $u_1(s) + u_2(s) = k$, $\forall s \in S_1 \times S_2$, $k \in \mathbb{R}$
- We can simplify TPZS by defining it as a **Matrix Game**

$$\begin{aligned}
&S_1 = \{s_{11}, s_{12}, ..., s_{1m}\} = \{1, 2, ..., m\} \\
&S_2 = \{s_{21}, s_{22}, ..., s_{2n}\} = \{1, 2, ..., n\} \\
&A = [a_{ij}] = [u_1(i,j)] \ \forall i \in S_1, \ \forall j \in S_2 \\
&u_2(i,j) = -a_{ij} \ \forall i \in S_1, \ \forall j \in S_2
\end{aligned}$$

- The Matching Pennies game can be rewritten as $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$
- Player 1 is called **Row Player** and Player 2 is called **Column Player**
- Suppose row player plays $i \in S_1$, then column player will play a strategy so as to $\min_{j \in S_2} a_{ij}$, now the row player looks to maximize this value

$$\max_{i \in S_1} \min_{j \in S_2} a_{ij}$$

  This is called **maxminimization** which is a consequence of rationality in matrix games.
- Suppose column player plays $j \in S_2$, then row player will play a strategy so as to $\max_{i \in S_1} a_{ij}$, now the column player looks to minimize this value

$$\min_{j \in S_2} \max_{i \in S_1} a_{ij}$$

  This is called **minmaximization**.
- Rationality of player 1 translates to maxminimization
- Rationality of player 2 translates to minmaximization
- Consider the game $\begin{bmatrix} 100 & 200 & 100 \\ 0 & -100 & 200 \\ -100 & 0 & -200 \end{bmatrix}$

$$\begin{aligned}
&\underline{v_1} = \max(100, -100, -200) = 100, \ \text{maxmin strategy} = 1 \\
&\overline{v_1} = \min(100, 200, 200) = 100, \ \text{minmax strategy against } 1 = 1 \\
&\underline{v_2} = \max(-100, -200, -200) = -100, \ \text{maxmin strategy} = 1 \\
&\overline{v_2} = \min(-100, 100, 200) = -100, \ \text{minmax strategy against } 2 = 1
\end{aligned}$$

  $\underline{v_1} = -\underline{v_2} = \underline{v} = \max_{i \in S_1} \min_{j \in S_2} a_{ij}$
  $\overline{v_1} = -\overline{v_2} = \overline{v} = \min_{j \in S_2} \max_{i \in S_1} a_{ij}$
- In general, we have seen that $\underline{v} \neq \overline{v}$, of course $\overline{v} \geq \underline{v}$
- If $\underline{v} = \overline{v}$, then the common number $v = \underline{v} = \overline{v}$ is called the **value of the game**.
- **Saddle Point of a Matrix Game**: In a matrix game A, a saddle point is any element a_{ij} which is simultaneously a column max and a row min

$$a_{ij} \geq a_{kj} \ \forall k = 1, 2, ..., m \text{ and } a_{ij} \leq a_{il} \ \forall l = 1, 2, ..., n$$

$$\text{Saddle Point} \equiv \text{PSNE}$$

  Suppose a_{ij} is a saddle point

$$\begin{aligned}
&\iff u_1(i,j) \geq u_1(i',j) \ \forall i' \in S_1 \text{ and } u_2(i,j) \geq u_2(i,j') \ \forall j' \in S_2 \\
&\iff (i,j) \text{ is a PSNE}
\end{aligned}$$

- A matrix game has a saddle point iff $\underline{v} = \overline{v}$, i.e., maxmin=minmax.

- If a_{ij} and a_{hk} are saddle points, then a_{ik} and a_{hj} are also saddle points.
- Saddle point(PSNE) is also robust to deviations by the other player.
- This happens because a saddle point is also a row minimum. When the other player deviates, the current player is not worse off.
- Now we take a look at mixed strategies
  $x = (x_1, x_2, ..., x_m)$ =mixed strategy of row player
  $y = (y_1, y_2, ..., y_n)$ =mixed strategy of column player

$$u_1(s, y) = -u_2(x, y) = \sum_{i=1}^{m}\sum_{j=1}^{n} x_i y_j a_{ij} = xAy \text{ (Vector Notation)}$$

- When row player plays x, they are assured of $\min_{y\in\Delta(S_2)} xAy$. the row player should therefore choose x such that $\max_{x\in\Delta(S_1)} \min_{y\in\Delta(S_2)} xAy$. Thus, maxminimization is the row players rational(optimal) strategy.
- Similarly, minmaximization, $\min_{y\in\Delta(S_2)} \max_{x\in\Delta(S_1)} xAy$, is column player's rational(optimal) strategy.
- We have already soon a Lemma which showed that max over mixed strategy is equal to max over pure strategies

$$\max_{x\in\Delta(S_1)} xAy = \max_{i\in S_1} \sum_{j=1}^{n} a_{ij} y_j \text{ and } \min_{y\in\Delta(S_2)} xAy = \min_{j\in S_2} \sum_{i=1}^{m} a_{ij} x_i$$

- Thus, the row players optimization problem becomes

$$\text{maximize } \min_j \sum_{i=1}^{m} a_{ij} x_i \text{ such that } x_i \geq 0, \sum_{i=1}^{m} x_i = 1 \qquad P_1$$

- Similarly, column player's optimization problem becomes

$$\text{minimize } \max_i \sum_{j=1}^{n} a_{ij} y_j \text{ such that } y_j \geq 0, \sum_{j=1}^{n} y_j = 1 \qquad P_2$$

- These can be converted to linear programming problems

$$\text{maxmize } z \text{ such that } z \leq \sum_{i=1}^{m} a_{ij} x_i \ \forall j = 1, 2, ..., n, \ x_i \geq 0 \ \forall i = 1, 2, ..., m \text{ and } \sum_{i=1}^{m} x_i = 1$$

- Similarly for column player

$$\text{maxmize } w \text{ such that } w \geq \sum_{j=1}^{n} a_{ij} y_j \ \forall i = 1, 2, ..., m, \ y_j \geq 0 \ \forall j = 1, 2, ..., n \text{ and } \sum_{j=1}^{n} y_j = 1$$

- Consider the game of rock, papers and scissors $\begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$

  LP1 in this case is, maximize z such that $z \leq x_2 - x_3, z \leq -x_1 + x_3, z \leq x_1 - x_2, \ x_1 + x_2 + x_3 = 1,$ and $x_1, x_2, x_3 \geq 0$

  LP2 in this case is, minimize w such that $w \geq -y_2 + y_3, w \geq y_1 - y_3, w \geq -y_1 + y_2, \ y_1 + y_2 + y_3 = 1,$ and $y_1, y_2, y_3 \geq 0$

  LP1 and LP2 are duals of each other.
  Solving the above equations, we get

$$x_1^* = x_2^* = x_3^* = y_1^* = y_2^* = y_3^* = \frac{1}{3}, \ z^* = w^* = 0$$

- **Strong Duality Theorem**: If an LP has an optimal solution, then its dual also has an optimal solution and moreover, the optimal values of the primal and dual are the same.

- **Minimax Theorem(Von Neumann and Oskar Morgenstern)**: Given a matrix game A, there exists $x^* = (x_1^*, x_2^*, ..., x_m^*)$, and $y^* = (y_1^*, y_2^*, ..., y_n^*)$ such that

$$\max_{x \in \Delta(S_1)} xAy^* = \min_{y \in \Delta(S_2)} x^*Ay$$

  Moreover, the profile (x^*, y^*) is a randomized saddle point(MSNE).
  Consider the primal LP, LP1 and its dual LP2.
  It is easy to prove that LP1 has an optimal solution. Suppose an optimal solution is given by $z^*, x_1^*, x_2^*, ..., x_m^*$
  LP2 also has an optimal solution and suppose an optimal solution is given by $w^*, y_1^*, y_2^*, ..., y_n^*$
  By strong duality theorem, the optimal values are the same, i.e., $z^* = w^*$
  We have $z^* = \sum_{i=1}^m a_{ij^*} x_i^*$ for some $j^* \in \{1, 2, ..., n\}$
  By feasibility $\sum_{i=1}^m a_{ij^*} x_i^* \leq \sum_{i=1}^m a_{ij} x_i^*$ for $j = 1, 2, ..., n$
  Therefore, $\sum_{i=1}^m a_{ij^*} x_i^* = \min_{j \in S_2} \sum_{i=1}^m a_{ij} x_i^* = \min_{y \in \Delta(S_2)} x^*Ay = z^*$
  Similarly, we can show $w^* = \max_{x \in \Delta(S_1)} xAy^*$
  Since, $z^* = w^* \implies$ we have proved $\min_{y \in \Delta(S_2)} x^*Ay = \max_{x \in \Delta(S_1)} xAy^*$

- Next we show (x^*, y^*) is a MSNE

$$u_1(x^*, y^*) = x^*Ay^* \geq \min_{y \in \Delta(S_2)} x^*Ay = \max_{x \in \Delta(S_1)} xAy^* \geq u_1(x, y^*) \ \forall x \in \Delta(S_1)$$

  Similarly,

$$u_2(x^*, y^*) = -x^*Ay^* \geq -\max_{x \in \Delta(S_1)} xAy^* = \min_{y \in \Delta(S_2)} x^*Ay \geq u_2(x^*, y) \ \forall y \in \Delta(S_2)$$

  Therefore, (x^*, y^*) is a MSNE

- Given a matrix game A, (x^*, y^*) is a MSNE iff

$$x^* \in \arg\max_{x \in \Delta(S_1)} \min_{y \in \Delta(S_2)} xAy$$

$$y^* \in \arg\min_{y \in \Delta(S_2)} \max_{x \in \Delta(S_1)} xAy$$

  Moreover,

$$u_1(x^*, y^*) = -u_2(x^*, y^*) = \max_{x \in \Delta(S_1)} \min_{y \in \Delta(S_2)} xAy = \min_{y \in \Delta(S_2)} \max_{x \in \Delta(S_1)} xAy$$

3 Cooperative Games

- **Correlated Strategies**: Consider the game $\prod = \langle N, (S_i), (u_i) \rangle$, we define a coalition of players $C \subseteq N$, and $S_C = \times_{i \in C} S_i$, $\tau_C \in \Delta(S_C)$ represents a probability distribution on the joint actions of players in coalition C.

- Consider a game with $N = \{1, 2, 3\}$, $S_1 = \{x_1, y_1\}$, $S_2\{x_2, y_2\}$, and $S_3 = \{x_3, y_3, z_3\}$.
  Consider the coalition $S_2 \times S_3 = \{(x_2, x_3), (x_2, y_3), (x_2, z_3), (y_2, x_3), (y_2, y_3), (y_2, z_3)\}$
  A correlated strategy of this coalition would be

$$\left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{12}, \frac{1}{12}, \frac{1}{12}\right) \text{ or } \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{32}\right)$$

- Suppose $\alpha \in \Delta(S_N)$, then we can define naturally $u_i(\alpha) = \sum_{s \in S} \alpha(s) u_i(s)$

- Expected payoff allocation to players when they play a correlated strategy, $u(\alpha) = (u_1(\alpha), ..., u_n(\alpha))$

- Note the difference between a correlated strategy and a mixed strategy profile.

$$\text{Correlated Strategy } \alpha \in \Delta(\times S_i), \text{ Mixed Strategy } \sigma \in \times(\Delta(S_i))$$

- Consider a game with $N = \{1, 2\}$, $S_1 = \{x_1, y_1\}$, and $S_2 = \{x_2, y_2\}$
  Suppose $\sigma = (\sigma_1, \sigma_2)$ with $\sigma_1 = (\delta_1, 1 - \delta_1)$, $\sigma_2 = (\delta_2, 1 - \delta_2)$
  Then, $u_i(\sigma) = \delta_1\delta_2 u_i(x_1, x_2) + (1 - \delta_1)\delta_2 u_i(y_1, x_2) + \delta_1(1 - \delta_2) u_i(x_1, y_2) + (1 - \delta_1)(1 - \delta_2) u_i(y_1, y_2)$
  This $u_i(\sigma)$ will be the same as $u_i(\alpha)$ where

$$\alpha = (\delta_1\delta_2, (1 - \delta_1)\delta_2, \delta_1(1 - \delta_2), (1 - \delta_1)(1 - \delta_2))$$

- Given any mixed strategy profile σ, we can find a correlated strategy α, such that $u_i(\alpha) = u_i(\sigma) \ \forall i \in N$

- The vice versa is not always possible thus correlated strategies produce a richer space of utility allocations.

- Correlated strategies are key to the study of cooperative games.

- We will see two classes of cooperative games, Games with contracts and Games with communication.

3.1 Games with Contracts

- Consider the modified prisoner's dilemma

1 \ 2	x_2	y_2
x_1	$2,2$	$0,6$
y_1	$6,0$	$1,1$

(a) Payoff Matrix

(y_1, y_2) is a SDSE and a unique NE
However, (x_1, x_2) yields a better payoff

Figure 4: Modified Prisoner's Dilemma

Rational players cannot achieve (x_1, x_2) in non-coop games
A contract can be as follows

1. If both sign, players play (x_1, x_2)
2. If only 1 signs, 1 plays y_1 (SDS)
3. If only 2 signs, 2 plays y_2 (SDS)
4. If neither signs, business as usual (non coop game)

We now expand the strategy sets as

$$S_1 = \{x_1, y_1, a_1\},\ a_1 : \text{Player 1 signs the contract}$$
$$S_2 = \{x_2, y_2, a_2\},\ a_2 : \text{Player 2 signs the contract}$$

1 \ 2	x_2	y_2	a_2
x_1	$2,2$	$0,6$	$0,6$
y_1	$6,0$	$1,1$	$1,1$
a_1	$6,0$	$1,1$	$2,2$

(a) Payoff Matrix

In this expanded game,
(a_1, a_2) is a WDSE
(y_1, y_2) is a PSNE

Figure 5: Modified Prisoner's Dilemma with Contract 1

Note that both the players signing the contract is a correlated strategy.
The players will be able to cooperate even better by signing one more contract

1. If both sign, they toss a coin and play (x_1, y_2) or (y_1, x_2)
2. If only 1 signs, player 1 plays y_1
3. If only 2 signs, player 2 plays y_2
4. If neither signs, business as usual with only contract a

We now expand the strategy sets as

$$S_1 = \{x_1, y_1, a_1, b_1\},\ b_1 : \text{Player 1 signs the contract}$$
$$S_2 = \{x_2, y_2, a_2, b_2\},\ b_2 : \text{Player 2 signs the contract}$$

1 \ 2	x_2	y_2	a_2	b_2
x_1	$2,2$	$0,6$	$0,6$	$0,6$
y_1	$6,0$	$1,1$	$1,1$	$1,1$
a_1	$6,0$	$1,1$	$2,2$	$1,1$
b_1	$6,0$	$1,1$	$1,1$	$3,3$

(a) Payoff Matrix

(b_1, b_2) is a PSNE with $(3,3)$
(a_1, a_2) is a PSNE with $(2,2)$
(y_1, y_2) is a PSNE with $(1,1)$
$\left(\left(0, 0, \frac{2}{3}, \frac{1}{3}\right), \left(0, 0, \frac{2}{3}, \frac{1}{3}\right)\right)$ is MSNE with $\left(\frac{5}{3}, \frac{5}{3}\right)$

Figure 6: Modified Prisoner's Dilemma with Contract 2

- Each contract can be represented by correlated strategies played by all coalitions.
  Contract 1: $\tau = (\tau_1, \tau_2, \tau_{12})$
  $\tau_1 : (x_1 : 0, y_1 : 1)$, and $\tau_2 : (x_2 : 0, y_2 : 1)$
  $\tau_{12} : ((x_1, x_2) : 1, (x_1, y_2) : 0, (y_1, x_2) : 0, (y_1, y_2) : 0)$
  Contract 2: $\tau = (\tau_1, \tau_2, \tau_{12})$
  $\tau_1 : (x_1 : 0, y_1 : 1)$, and $\tau_2 : (x_2 : 0, y_2 : 1)$
  $\tau_{12} : ((x_1, x_2) : 0, (x_1, y_2) : 0.5, (y_1, x_2) : 0.5, (y_1, y_2) : 0)$

- A contact can therefore be defined as $\tau = (\tau_C)_{C \subseteq N}$, $C \neq \phi$
  $\tau_C \in \Delta(S_C)$ is the correlated strategy that would be implemented iff the players in C have signed the contract.

- **Contract Signing Game**: A game in which signing a contract is included as an additional strategy.

- All players would be willing to sign a contract when $(a_1, ..., a_n)$ is a NE of the contract signing game.

- **Individually Rational Correlated Strategy**: $\alpha \in \Delta(S)$ is said to IR if $u_i(\alpha) \geq v_i \ \forall i \in N$ where v_i is an acceptable lower bound

$$\underline{v_i} = \max_{\tau_i \in \Delta(S_i)} \min_{\tau_{-i} \in \Delta(S_{-i})} u_i(\tau_i, \tau_{-i})$$
$$\overline{v_i} = \min_{\tau_{-i} \in \Delta(S_{-i})} \max_{\tau_i \in \Delta(S_i)} u_i(\tau_i, \tau_{-i})$$

  It can be shown that $\underline{v_i} = \overline{v_i} = v_i$. In fact,

$$\underline{v_i} = \max_{\tau_i \in \Delta(S_i)} \min_{s_{-i} \in S_{-i}} u_i(\tau_i, s_{-i}) \text{ and } \overline{v_i} = \min_{\tau_{-i} \in \Delta(S_{-i})} \max_{s_i \in S_i} u_i(s_i, \tau_{-i})$$

- We will now prove two results that will establish the connection between IR of correlated strategy and NE of contract signing games.

Result 1 Suppose α is IR, define contract $\tau = (\tau_C)_{C \subseteq N}$, $\tau_N = \alpha$
$\tau_i =$ minmax correlated strategy of $-i$ against player i
τ_C for any other C, arbitrary.
Suppose a_i denotes that player i signs contract

$$u_i(a_1, a_2, ..., a_n) = u_i(\alpha) \geq v_i$$

Consider $u_i(s_i, a_{-i}) = u_i(s_i, \tau_{-i}) \leq v_i\{$ since τ_{-i} is minimax against $i\}$
Thus, $u_i(a_i, a_{-i}) \geq u_i(s_i, a_{-i}) \ \forall s_i \in S_i, \ \forall i \in N$
Hence, $(a_1, ..., a_n)$ is a PSNE.
If α is an IR correlated strategy, we can find a contract such that all players signing the contract is a NE. Players would be willing to sign such a contract.

Result 2 Suppose we are given a contract τ such that $(a_1, ..., a_n)$ is a NE
Can $u_i(a_1, ..., a_n) < v_i$ for any $i \in N$?
Suppose the above is true. Then choose for player i the maxmin strategy(say i). Since, v_i is the maxmin value of the player i,

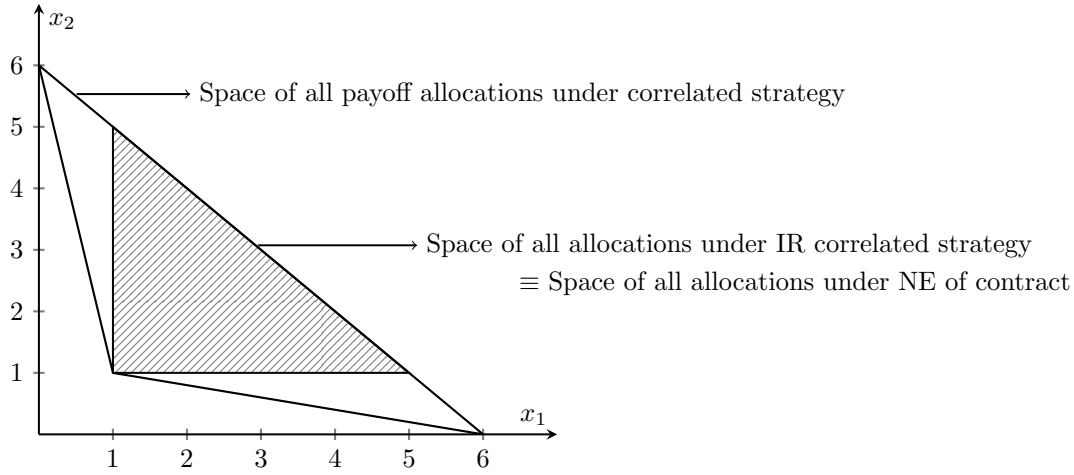
$$u_i(s_i, a_{-i}) \geq v_i$$

However, (a_i, a_{-i}) being a NE,

$$u_i(a_, a_{-i}) \geq u_i(s_i, a_{-i}) \geq v_i \text{ (Contradiction)}$$

Therefore, $u_i(a_1, ..., a_n) \geq v_i, \ \forall i \in N$
The correlated strategy corresponding to all players signing the contract is therefore IR.

- Given a contract $\tau = (\tau_C)_{C \subseteq N}$, if all players signing the contract $(a_1, ..., a_n)$ is a NE then τ_N is an IR correlated strategy



- Both the spaces above are convex, closed, and bounded.

3.2 Games with Communication

- **Correlated Equilibrium**: This is a correlated strategy $\alpha \in \Delta(S)$ which when proposed by a trusted mediator is acceptable to all the (rational) players, the players are happy to play the strategy recommended by the mediator .
- PSNE(MSNE) is a pure strategy profile(mixed strategy profile) which when proposed by a trusted mediator is acceptable to all the (rational) players.
- Consider the following game with communication

1 \ 2	x_2	y_2
x_1	$5, 1$	$0, 0$
y_1	$4, 4$	$1, 5$

(a) Payoff Matrix

Game has 3 NE
$(x_1, x_2), (y_1, y_2)$ yielding payoffs $(5, 1)$ and $(1, 5)$
$\left(\left(\frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, \frac{1}{2}\right)\right)$ yielding payoff $\left(\frac{5}{2}, \frac{5}{2}\right)$

Figure 7: Game with Communication

(y_1, x_2) is desirable, but it's not a NE.
Can be realized using a contract. But what to do if contracts are not allowed?
Can we recommend a correlated strategy which yields good payoffs and which the players are willing to follow under assumption of rationality and intelligence.

Correlated Strategy 1 $\alpha = ((x_1, x_2) : 0.5, (x_1, y_2) : 0, (y_1, x_2) : 0, (y_1, y_2) : 0.5)$
Here, $u_1(\alpha) = u_2(\alpha) = 3$
Mediator recommends (x_1, x_2) or (y_1, y_2) with equal probability. This correlated strategy is known to players.
Player 1's logic: x_1 is recommended $\rightarrow$ thinks 2 is recommended $x_2 \rightarrow$ plays x_1, thinking 2 plays x_2
Similar for y_1 and player 2

Correlated Strategy 2 $\alpha = ((x_1, x_2) : \frac{1}{3}, (x_1, y_2) : 0, (y_1, x_2) : \frac{1}{3}, (y_1, y_2) : \frac{1}{3})$
Here, $u_1(\alpha) = u_2(\alpha) = \frac{10}{3}$
Player 1's logic
$x_1 \rightarrow 1$ thinks 2 is recommended x_2, plays x_1 thinking 2 plays x_2
$y_1 \rightarrow$ thinks 2 is recommended x_2 with probability 0.5 and y_2 with probability 0.5, thinks 2 will play mixed strategy$(x_2 : 0.5, y_2 : 0.5)$, computes payoff as $\frac{5}{2}$ by playing x_2 and again $\frac{5}{2}$ by playing y_2. So, happy to play y_1.
Similar logic for player 2.

Correlated Strategy 3 $\alpha = ((x_1, x_2) : 0, (x_1, y_2) : 0, (y_1, x_2) : 1, (y_1, y_2) : 0)$
Here, $u_1(\alpha) = u_2(\alpha) = 4$
Player 1's logic
$y_1 \rightarrow 1$ knows 2 is recommended x_2, 1 plays x_1 since x_1 fetches 5 while y_1 only fetches 4.

This means player 1 does not obey the recommendation.
Similar logic for player 2.

Correlated strategies 1 and 2 are "correlated equilibria"
Correlated strategy 3 is not a correlated equilibrium.

- **Setup for Correlated Equilibrium**: Consider the game $\Pi = \langle N, (S_i), (u_i) \rangle$
  Mediator announces to all players a correlated strategy $\alpha \in \Delta(S)$ (Common Knowledge)
  Mediator picks a pure strategy profile $s \in S$, reveals to each player $i \in N$, the strategy s_i but not s_{-i}.
  Player i either obeys to play strategy s_i or chooses any other strategy according to $\delta_i : S_i \to S_i$.
  $\delta_i(s_i) =$ strategy chosen by player i when the mediator recommends s_i
  $\delta_i(s_i) = s_i$ means player i "obeys".
  $\alpha \in \Delta(S)$ is called a correlated equilibrium if

$$u_i(\alpha) \geq \sum_{(s_i, s_{-i}) \in S} \alpha(s_i, s_{-i}) u_i(\delta_i(s_i) s_{-i}) \ \forall \delta_i(s_i) : S_i \to S_i, \ \forall i \in N$$

- A correlated equilibrium is any correlated strategy which can be implemented self enforcingly by a mediator who makes non-binding recommendations. The above equations is equivalent to

$$\sum_{s_{-i} \in S_{-i}} \alpha(s_i, s_{-i}) \{u_i(s_i, s_{-i}) - u_i(s'_i, s_{-i})\} \geq 0 \ \forall s_i \in S_i, \ s'_i \in S_i, \ \forall i \in N$$

- Note that a correlated equilibrium α is a feasible solution of the following LP

$$\max 1 \text{ such that above equation satisfies and } , \alpha(s) \geq 0, \ \forall s \in S, \ \sum_{s \in S} \alpha(s) = 1$$

- If we want to find a welfare maximizing correlated equilibrium, we need to solve the above LP but with $\max 1$ replaced with $\max \sum_{i \in N} u_i(\alpha)$

- Consider the previous graph, a smaller region within it represents allocations under correlated equilibria. All these sets are convex, closed and bounded.

- Cooperative Game Theory looks at all possibilities and comes up with solution concepts that describe logical final outcomes of a cooperative game.

- Solution for Game Theory can be of **Set Type** (Core, Kernel, Bargaining set), and **Point Type** (Nash Bargaining solution, Shaley Value, Nucleolus)

- **Transferrable Utility (TU)**: Games in coalition/characteristic form.
  Defined by (N, v), where $N = \{1, 2, ..., N\}$ and $v : 2^N \to \mathbb{R}$
  $v(\phi) = 0$, $v(C) =$ "worth" of coalition C, $C \subseteq N =$ Total value that can be realized by the players in C.

- **House Allocation Game**
  Seller 1 has a house that they want to sell, they value the house at 1 Crore. There are 2 prospective buyers each having 2 Crore cash, $(2, 3)$.
  Seller 1 expects at least 1 Cr, each buyer can pay at most 2 Cr. Rule out the possiblity that both buyers buy.
  Suppose p is the sold value $1 \leq p \leq 2$
  $v(\phi) = 0$, $v(1) = 1$, $v(2) = v(3) = 2$ (Sale has not happened yet)
  $v(12) = p(\text{Seller gets}) + 2(\text{buyer got house}) + (2 - p)(\text{leftover}) = 4$
  $v(13) = p + 2 + (2 - p) = 4$
  $v(23) = 2 + 2 = 4$
  $v(123) = [p + 2 + (2 - p)](\text{sold to 1 player}) + 2 = 6$
  Trading will only happen when willingness to buy $\geq$ willingness to sell

- **Glove Market**
  $N_L = \{1, 2\}$, $N_R = \{3, 4, 5\}$, $N = \{1, 2, 3, 4, 5\}$
  $v(C) =$ Number of matched pairs $= \min(|N_L \cap C|, |N_R \cap C|)$

- **Blocking of allocation by coalition**: A coalition C blocks an allocation $(x_1, ..., x_n)$ if $\sum_{j \in C} x_j < v(C)$.
  We can always get another allocation $(y_1, ..., y_n)$ such thath $y_j > x_j \ \forall j \in C$ and $\sum_{j \in C} y_j = v(C)$.

3.3 Nash Bargaining Solution

- **Two Person Bargaining Problem**: Two players have the possibility of concluding a mutually beneficial agreement. There is a conflict about which agreement to conclude. No agreement may be imposed without the approval of players.

- The solution should depend only on
  the payoffs they would expect if the negotiations were to fail
  the payoffs that are feasible in the process of negotiation.

- **Applications**: Management(Labour arbitration), International bilateral relations, Duopoly markets/market design, Property settlement disputes, supply chain contracts, Property settlement, Negotiation protocols in multiagent systems.

- **Problem**: (F, v)
  F: feasible set $\subseteq \mathbb{R}^2$, is closed and convex
  $v : (v_1, v_2) \in \mathbb{R}^2$, **Disagreement point aka Default point**; status quo point solution that will be implemented in case no agreement is reached.
  $F \cap \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \geq v_1, x_2 \geq v_2\}$ is non-empty and bounded.

- Convexity of F: If $(x_1, x_2), (y_1, y_2) \in F$ then, for any $0 \leq \lambda \leq 1$,

  $$\lambda(x_1, x_2) + (1 - \lambda)(y_1, y_2) \in F$$

  this is a natural requirement since the convex combination can be achieved by the correlated strategy, $((x_1, x_2) : \lambda, (y_1, y_2) : (1 - \lambda))$

- F is closed: The limit of a sequence of feasible allocations is also feasible. It will be odd if the limit lies outside the feasible set.

- Typical choices of F
  $F = \{(u_1(\alpha), u_2(\alpha)) : \alpha \in \Delta(S_1 \times S_2)\}$
  $F = \{(u_1(\alpha), u_2(\alpha)) : \alpha \in \Delta(S_1 \times S_2)$ and α is IR$\}$
  $F = \{(u_1(\alpha), u_2(\alpha)) : \alpha$ is correlated equilibrium$\}$

- Disagreement point
  $v_i =$ maxmin value of player i (security value) (no regret)
  $v_i =$ Utility of player i under a focal Nash Equilibrium
  $v_i =$ Utility of player i under a rational threat equilibrium

- **Nash Axioms**: Nash brilliantly proposed five axioms that a solution to the bargaining problem should ideally satisfy, and showed that such a solution exists and is unique. Suppose a candidate solution is $f(F, v) = (f_1(F, v), f_2(F, v)) \in \mathbb{R}^2$

  1. **Individual Rationality**

     $$f_1(F, v) \geq v_1 \text{ and } f_2(F, v) \geq v_2$$

  2. **Strong Efficiency Axiom**
     $(x_1, x_2) \in F$ is strongly(Pareto) efficient if $\nexists y = (y_1, y_2) \in F$ such that $(y_1 \geq x_2, y_2 \geq x_2)$ and $(y_1 > x_1$, or $y_2 > x_2)$
     Strong efficiency axiom says that $f(F, v) \in F$ and is strongly efficient. This means there exists no other feasible solution that is strictly better for one player and not worse for the other player.

  3. **Scale Co-variance**
     For any $\lambda_1, \lambda_2, \mu_1, \mu_2 \in \mathbb{R}$ with $\lambda_1, \lambda_2 > 0$, define
     $G = \{(\lambda_1 x_1 + \mu_1, \lambda_2 x_2 + \mu_2) : (x_1, x_2) \in F\}$, and $\omega = (\lambda_1 v_1 + \mu_1, \lambda_2 v_2 + \mu_2)$
     then the solution will also follow the same scaling

     $$f(G, \omega) = (\lambda_1 f_1(F, v) + \mu_1, \lambda_2 f_2(F, v) + \mu_2)$$

  4. **Independence of Irrelevant Alternatives**
     If G is closed and convex, $G \subseteq F$ and $f(F, v) \in G$

     $$\implies f(G, v) = f(F, v)$$

  5. **Symmetry**
     Given, $\{(x_2, x_1) : (x_1, x_2) \in F\} = F$ and $v_1 = v_2$

     $$\implies f_1(F, v) = f_2(F, v)$$

- There exists a unique function f that satisfies all these axioms

$$f(F,v) = \underset{(x_1,x_2)\in F}{\arg\max}(x_1 - v_1)(x_2 - v_2) \text{ such that } x_1 \geq v_1, \text{ and } x_2 \geq v_2$$

- **Principle of Equal Gains**: This means $x_1 - v_1 = x_2 - v_2$. The two players should get the same gain over the default value. Lead to an **Egalitarian solution**.

- **Principle of Greatest Good**: Select an allocation that maximizes $x_1 + x_2$ (social welfare). Leads to an **Utilitarian solution**, **Welfare Maximizing solution**.

- Given a two-person bargaining problem (F, v), the egalitarian solution is the unique point $(x_1, x_2) \in F$ that is weakly efficient in F and satisfies $x_1 - v_1 = x_2 - v_2$.
  (x_1, x_2) is weakly efficient if $\nexists (y_1, y_2) \in F$ such that

  $$y_1 > x_1 \text{ and } y_2 > x_2$$

  A utilitarian solution is any allocation $(x_1, x_2) \in F$ such that

  $$x_1 + x_2 = \max_{(y_1,y_2)\in F} y_1 + y_2$$

  Both of these violate Scale covariance.

- **λ-Egalitarian solution**: Given $\lambda_1, \lambda_2 > 0$, $\mu_1, \mu_2 \in \mathbb{R}$, define for any $y = (y_1, y_2) \in \mathbb{R}^2$

  $$L(y) = (\lambda_1 y_1 + \mu_1, \lambda_2 y_2 + \mu_2)$$

  Given the problem (F, v), define

  $$L(F) = \{L(y) : y \in F\}$$

  The egalitarian solution of $(L(F), L(v))$ will be $L(x)$ where x is the unique weakly efficient point in F such that

  $$\lambda_1(x_1 - v_1) = \lambda_2(x_2 - v_2)$$

  If $\lambda_1 = \lambda_2 = 1$, the solution is said to be simple egalitarian.

- **λ-Utilitarian solution**: Taking the same definition as above
  The utilitarian solution of $(L(F), L(v))$ will be $L(x)$ where x is the unique weakly efficient point in F such that

  $$\lambda_1 x_1 + \lambda_2 x_2 = \max_{(y_1,y_2)\in L(F)} (\lambda_1 y_1 + \lambda_2 y_2)$$

  If $\lambda_1 = \lambda_2 = 1$, the solution is said to be simple utilitarian.

- If λ_1 increases and λ_2 decreases, the λ-egalitarian solutions will lead to an IR, weakly efficient frontier moving in the direction of decreasing payoff to player 1. The λ-utilitarian solutions will lead to a weakly efficient frontier, moving in the direction of increasing payoff to player 1.

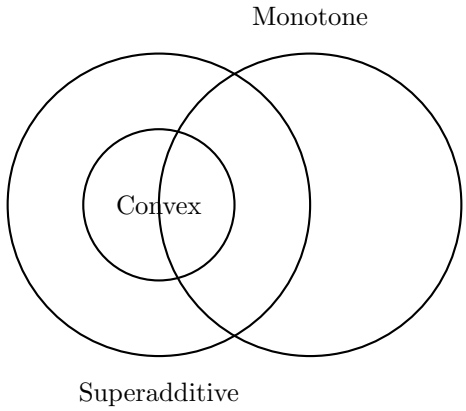
- If the bargaining problem (F, v) is such that the solution is $> v$, then $\exists \lambda = (\lambda_1^*, \lambda_2^*) > (0, 0)$ such that the λ-egalitarian solution coincides with the λ-utilitarian solution.
  This λ_1^*, λ_2^* is called **natural scale factor**.
  This solution will also coincide with the Nash Bargaining solution.

- NBS can be viewed as a natural synthesis of the principles of equal gain and greatest good(welfare maximization).

- **Monotone Games**: $v(C) \leq v(D)\ \forall C \subseteq D \subseteq N$

- **Super additive games**: $v(C \cup D) \geq v(C) + v(D)\ \forall C, D \subseteq N, C \cap D = \phi$. Union of non-overlapping coalitions produce no less value than the individual coalitions combined.

- **Convex Games**: $v(C \cup D) \geq v(C) + v(D) - v(C \cap D)\ \forall C, D \subseteq N$
  $v(C \cup \{i\}) - v(C) \leq v(D \cup \{i\}) - v(D)\ \forall C \subseteq D \subseteq N, \forall i \in N$
  Marginal contribution of a new player to a larger coalition is larger than that of the player for a smaller coalition.



3.4 The Core

- **The core** is a set solution concept for coalitional games. The notion of "core" is motivated by the consideration to keep all coalitions "happy" and "stable".
- Consider a TU game, (N, v)
  $x = (x_1, ..., x_n) \in \mathbb{R}^n$, an allocation is called **coalitionally rational** if $\sum_{i \in C} x_i \geq v(C)\ \forall C \subseteq N$.
- Special case of coalitional rationality is individual rationality, which means $x_i \geq v(\{i\})\ \forall i \in N$
- The allocation is called **collectively rational** if $\sum_{i \in N} x_i = v(N)$
- The core of (N, v) is the collection of all collectively rational and coalitionally rational allocations

$$C(N, v) = \{(x_1, ..., x_n) \in \mathbb{R}^n : \sum_{i \in C} x_i \geq v(C)\ \forall C \subseteq N \text{ and } \sum_{i \in N} x_i = v(N)\}$$

- A coalition C is said to "block" an allocation $(x_1, ..., x_n)$ if

$$\sum_{i \in C} x_i < v(C)$$

  Note that we can "improve upon" this allocation with (uncountably infinite number of) allocation $(y_1, ..., y_n)$ such that

$$\sum_{i \in C} y_i \leq v(C) \text{ and } y_i > x_i\ \forall i \in N$$

- The core $C(N, v)$ is the set of all allocations $x \in \mathbb{R}^n$ such that no coalition can block x.
- A core allocation is stable and arises out of "effective" negotiations.
- If a "referee" recommends a core allocation, no coalition will be unhappy about it.
- $C(N, v)$ is convex, closed and bounded.
- $C(N, v)$ can be empty, singleton, or uncountably infinite.
- **Shapley-Bondereva Characterization**: Consider the following minimization problem,

$$\min x_1 + ...x_n \text{ such that } \sum_{i \in C} x_i \geq v(C)\ \forall C \subseteq N,\ x_i \in \mathbb{R},\ \forall i \in N$$

  This LP certainly has a feasible solution. An optimal solution will give the least amount of transferrable utility that is necessary for no coalition to block it.
  Suppose $(x_1^*, ..., x_n^*)$ is an optimal solution

$$x_1^* + ... + x_n^* \geq v(N)$$

$x_1^* + ... + x_n^* > v(N)$	$x_1^* + ...x_n^* = v(N)$
The core is empty	The core is non-empty and each optimal solution is an element

- **Core of a Convex Game**: Suppose (N, v) is convex, then $x = (x_1, ..., x_n) \in C(N, v)$ where $x_1 = v(1)$, $x_2 = v(12) - v(1)$, $x_3 = v(123) - v(12)$, ..., $x_n = v(N) - v(N \backslash \{n\})$
  If we allocate marginal contributions to players in a convex game, the allocation is in the core.

- Suppose Π is any permutation of N and let
  $P(\Pi, i)$ = Set of players preceding i in Π
  Define $m(\Pi, i) = v(P(\Pi, i) \cup \{i\}) - v(P(\Pi, i))$ then $(x_1, ..., x_n) \in C(N, v)$ where

$$x_1 = m(\Pi, 1), x_2 = m(\Pi, 2), ..., x_n = m(\Pi, n)$$

- $x_1 + x_2 + ... + x_n = v(N)$, so above is collectively rational.
  To prove coalitional rationality, show that $\sum_{i \in E} x_i \geq v(E)\ \forall E \subseteq N$. We can write $E = \{i_1, i_2, ..., i_k\}$ for some $k \leq n$.
  WLoG assume $i_1 < i_2 < ... < i_k$, we observe that $i_{j-1} \leq i_j - 1$ and therefore

$$\{i_1, i_2, ..., i_{j-1}\} \subseteq \{1, 2, ..., i_j - 1\}\ \forall j \in \{1, 2, ..., k\}$$

  Recall the definition of a convex game

$$v(D \cup \{z\}) \geq v(C \cup \{z\})\ \forall C \subseteq D \subseteq N\ \forall z \in N \backslash D$$

  Choose $C = \{i_1, ..., i_{j-1}\}$, $D = \{1, 2, ..., i_j - 1\}$, and $z = i_j \in N \backslash D$
  We obtain

$$v(\{1, 2, ..., i_j - 1, i_j\}) - v(\{1, 2, ..., i_j - 1\}) \geq v(\{i_1, i_2, ..., i_{j-1}, i_j\}) - v(\{i_1, i_2, ..., i_{j-i}\})$$

  Next, we observe that

$$\begin{aligned}
&x_{i_1} = v(\{1, 2, ..., i_1\}) - v(\{1, 2, ..., i_1 - 1\}) \geq v(\{i_1\}) - v(\phi) \\
&x_{i_2} = v(\{1, 2, ..., i_2\}) - v(\{1, 2, ..., i_2 - 1\}) \geq v(\{i_1, i_2\}) - v(\{i_1\}) \\
&\vdots \\
&x_{i_k} = v(\{1, 2, ..., i_k\}) - v(\{1, 2, ..., i_k - 1\}) \geq v(\{i_1, ..., i_k\}) - v(\{i_1, ..., i_{k-1}\}) \\
&x_{i_1} + ...x_{i_k} \geq v(\{i_1, ..., i_k\}) = v(E)
\end{aligned}$$

  Clearly, the above is true $\forall E \subseteq N$ and thus proves coalitional rationality.

3.5 Shapley Value

- Given a TU game, **the shapley value** $\varphi(N, v)$ provides a reasonable and fair way of dividing gains from cooperation given the strategic dynamics captured by the characteristic function v.

- Uses "marginal contribution" as the guiding principle.

- Based on three axioms
  $\varphi(N, v) = (\varphi_1(N, v), ..., \varphi_n(N, v))$, we will use $\varphi(v) = (\varphi_1(v), ..., \varphi_n(v))$

  1. **Symmetry**
     Suppose $\Pi : N \to N$ is a permutation of $(1, 2, ...n)$. Define the game $(N, \Pi v)$ as follows

$$\Pi v(C) = v(\Pi^{-1}(C))$$

     The symmetry axiom states $\varphi_{\Pi i}(\Pi v) = \varphi_i(v)$, another way to state $\varphi_i(\Pi v) = \varphi_{\Pi^{-1} i}(v)$
     This basically asserts that "renaming" does not affect the shapely value of a player.

  2. **Linearity**
     Given two games (N, v) and (N, ω), consider the convex combination game with characteristic function

$$pv + (1 - p)\omega \text{ where } p \in [0, 1]$$

     The axiom states

$$\varphi_i(pv + (1 - p)\omega) = p\varphi_i(v) + (1 - p)\varphi_i(\omega)\ \forall i \in N\ \forall p \in [0, 1]$$

3. **Carrier**
   $D \subseteq N$ is called a "carrier" if $v(C \cap D) = v(C)\ \forall C \subseteq N$
   Suppose D is a carrier, then the following are true
   Suppose $i \notin D$, then
   $$v(\{i\}) = v(\{i\} \cap D) = v(\phi) = 0$$
   Thus, an element not belonging to D is a "dummy" player. This means all influential players are contained in D. D may contain non-influential players also.
   If D is a carrier, then $D \cup \{i\}$ is also a carrier for any $i \in N$
   $$v(C \cap (D\{i\})) = v((C \cap D) \cup (C \cap \{i\})) = v(C \cap D) = v(C)$$
   N is always a carrier, since $v(C \cap N) = v(C)$
   Suppose D is a carrier, then $v(N \cap D) = v(N)$. This means $v(D) = v(N)$ for any carrier D.
   The axiom says that if D is a carrier then
   $$\sum_{i \in D} \varphi_i(v) = v(D) = v(N)$$
   This means that the value of the grand coalition is the sum of shapley value of elements in any carrier D.
   Since, N is a carrier, $\sum_{i \in N} \varphi_i(v) = v(N)$
   $$\varphi_i(v) = 0\ \forall i \notin D$$

- **Shapley Theorem**: There exists exactly one mapping $\varphi : \mathbb{R}^{2^n - 1} \to \mathbb{R}^n$ that satisfies all the axioms and
  $$\varphi_i(v) = \sum_{C \subseteq N - i} \frac{|C|!(n - |C| - 1)!}{n!} \{v(C \cup \{i\}) - v(C)\}$$
  expected marginal contribution of player i to the worth of any coalition.
  The factorial term is the probability that in any permutation of $(1, 2, ..., n)$, the members of C are ahead of player i.

- **Alternate Formulation**: Suppose $\Pi : N \to N$ is a permutation
  $P(\Pi, i)$ = Set of players preceding i in permutation Π
  $m(\Pi, i)$ = Marginal contribution of i to the set of predecessors in $\Pi = v(P(\Pi, i) \cup \{i\}) - v(P(\Pi, i))$
  It can be shown that
  $$\varphi_i(v) = \frac{1}{n!} \sum_{\pi \in \Pi(N)} m(\pi, i) \text{ for } i = 1, 2, ..., n$$
  Expected marginal contribution of i to its predecessors with expectation over all permutations.

- **Shapley value in a convex game**: We have seen for any permutation Π that $(m(\Pi, 1), ..., m(\Pi, n)) \in C(N, v)$. Since, the core is a convex set, we therefore know that any convex combination of core elements is in the core. Consider the following convex combination
  $$\sum_{\pi \in \Pi(N)} \frac{1}{n!} m(\pi, i)$$
  This is precisely the Shapley value. Hence, in convex games $\varphi_i(v) \in C(N, v)$

- **Applications of Shapley Value**
  1. Influence maximization in social networks, influence limitations in rumor networks etc.
  2. Vaccination planning
  3. Cost sharing, Surplus sharing
  4. SHAP(Shapley Additive Exploitations): An approach based on Shapley value to explain the output of an ML model.

3.6 Nucleolus

- **Excess**: Given a TU game (N, v), a coalition C, and an allocation $x = (x_1, ..., x_n)$ the excess of C at x is defined as
$$e(C, x) = v(C) - \sum_{i \in C} x_i$$
The excess $e(C, x)$ is the **net transferable utility** that the coalition C would be left with after allocating x_i to player i where $i \in C$.
- If $e(C, x) \geq 0$, the implication is that the coalition, on its own, would be able to achieve its share of the allocation x.
- The excess of a coalition C w.r.t x is a measure of unhappiness of C with allocation x.
- **Domination**: An imputation $x = (x_1, ..., x_n)$ is said to dominate another imputation $y = (y_1, ..., y_n)$ if there is some coalition $C \subseteq N$ such that
$$e(C, x) \geq 0 \text{ and } x_i > y_i \ \forall i \in C$$
- An imputation $x = (x_1, ..., x_n)$ is said to be **undominated** if no other imputation dominates it.
- Note that the core of a TU game is simply the collection of all undominated imputations.
- **Internal Stability**: Given a TU game (N, v), a set of imputations Z is said to be internally stable if
$$\forall x, y \in Z, \ \forall C \subseteq N, x_i > y_i \ \forall i \in C \implies e(C, x) < 0$$
This means no imputation in Z is dominated by any other imputation in Z.
- Internal stability implies that if an imputation y belonging to Z is proposed, then no coalition C can block this since a strictly better, feasible outcome does not exist in Z for the players in C.
- **External Stability**: Given a TU game (N, v)), a set of imputations Z is said to be externally stable if every imputation not in Z is dominated by some imputation in Z. For all imputations y of $(N, v), y \notin Z \implies$ there exists an $x \in Z$ such that there is at least one coalition $C \subseteq N$ for which
$$x_i > y_i \ \forall i \in C \implies e(C, x) \geq 0$$
- External stability implies that if an imputation y not belonging to Z is proposed, then there is at least one coalition C of players that could block the adoption of y by insisting on getting their share of some strictly better outcome in Z that is feasible for them.
- **Stable Set (Von-Neumann-Morgenstern solution)**: A stable set of a TU game (N, v) is a set of imputations Z satisfying internal stability and external stability.
- The nucleolus, which is also a unique allocation, is primarily based on bargaining considerations.
- Let us say we find an imputation that minimizes the largest among all excesses $e(C, x)$. This means the chosen imputation makes the most unhappy coalition as little unhappy as possible.
- **The nucleolus** is based on this idea and minimizes not only the level of unhappiness of the most unhappy coalition but also the levels of unhappiness of the second most unhappy coalition, third most unhappy coalition, etc.
- Consider a three player game with $N = \{1, 2, 3\}$ and characteristic function defined by
$$v(1) = v(2) = v(3) = 0; v(12) = 60; v(13) = 80; v(23) = 100; v(123) = 105$$
Given an allocation $x = (20, 35, 50)$, the excess values for the coalitions are
$$e(1, x) = -20; e(2, x) = -35; e(3, x) = -50; e(12, x) = 5; e(13, x) = 10; e(23, x) = 15; e(123, x) = 0$$
The largest excess is $e(23, x) = 15$. We can reduce this by trying to allocate $y = (15, 35, 55)$
$$e(12, y) = 10; e(13, y) = 10; e(23, y) = 10$$
If we try to lower this any further, it would raise at least one other excess beyond 10. Thus, the allocation y achieves the lowest excess for any coalition.

- In general, there may be multiple imputations which minimize the second-largest excess to obtain a subset of the above set of imputations. Next, we minimize the third-largest excess, fourth-largest excess, etc. until we end up with a unique imputation.
- This unique imputation is called **the nucleolus**.
- Consider any allocation, $x = (x_1, ..., x_n)$ and let $e_k(x)$ be the largest excess generated by any coalition with the allocation x. This means

$$|\{C \subseteq N : e(C, x) \geq e_k(x)\}| \geq k$$
$$|\{C \subseteq N : e(C, x) > e_k(x)\}| < k$$

  Now, x belongs to Core (N, v) implies $e_1(x) \leq 0$. Suppose we denote by $J(k)$ the set of all imputations that minimize e_k where e_k is the kth largest excess and by $I(N, v)$ the set of all imputations of (N, v). Then

$$J(1) = \underset{x \in I(N,v)}{\arg\min}\, e_1(x)$$

  If the core is non-empty, then $J(1) \subseteq$ core (N, v). We can define $J(2), J(3), ..., J(2^{|N|} - 1)$ as follows

$$J(k) = \underset{x \in J(k-1)}{\arg\min}\, e_k(x) \text{ for } k = 2, 3, ..., 2^{|N|-1}$$

- It was shown by Schmeidler that the set $J(2^{|N|-1})$ is a singleton, and this point is called **the nucleolus** of the TU game (N, v).
- Note that when the core is non-empty, for any imputations in the core, all the excesses are zero or negative.
- For finding the nucleolus, we choose the imputation in the core which makes the least negative excess as negative as possible
- Geometrically, the nucleolus is a point in the core which is as far inside the core as possible.
- The nucleolus always exists and is unique.
- If the core is non-empty, the nucleolus belongs to the core.
- The nucleolus always belongs to the bargaining set for the grand coalition. Also, it always belongs to the kernel.
- The nucleolus is not necessarily equal to the Shapley value. In fact, the nucleolus may allocate the surplus to players in a ruthless manner, while Shapley value is built on the principle of fairness.
- The nucleolus can be computed by solving a series of linear programs.

4 Mechanism Design

- In game theory, we inherit a game and are required to reason about it–a lot of economics if concerned with taking existing economic institutions and trying to explain/predict the outcomes that those institutions generate. This is called the **positive economics**.
- By contrast, in mechanism design, we start by identifying desirable outcome and work backwards to figure out what institutions would generate those outcomes.
- Mechanism design is inverse game theory, it is the engineering side of game theory where we get to design a game—design rules—such that even with strategic (self-interested) participants a desired outcome is obtained.
- Mechanism design is the science of rule making.
- What procedures can we design such that even with self-interested agents (strategic participants) the system yields a socially optimal outcome?
- Most situations of interest have **private information**. When we design a game, we do not design it in vacuum. There are parts of the system that the designer does not control, e.g., the participants and their inherent preferences.
- The presence of strategic participants who hold part of the "input" (private information) is what separates mechanism design from algorithm design.
- Tim Roughgarden's Lecture Notes: `https://theory.stanford.edu/~tim/f13/l/f13.pdf`

4.1 Single Item Auctions

- 1 good and n strategic bidders. We need a framework to quantitatively understand the behavior of the bidders. That is, we need to model the bidders using game theoretic notions.
- **Valuation**: Each bidder i has valuation v_i, the maximum amount that i is willing to pay for the item. Thus, the bidder wished to acquire the item as cheaply as possible, provided its selling price is at most v_i.
- An important **assumption** is that the valuations are **private**. v_i is known only to the agent i, it is unknown to the auctioneer and the other agents.
- **Quasilinear utility model**

$$u_i = \begin{cases} 0, & \text{if } i \text{ looses the auction} \\ v_i - p, & \text{if } i \text{ wins at sale price } p \end{cases}$$

- We will model the bidders as acting to maximize their quasilinear utility.
- **Sealed Bid Auctions**
  1. Each bidder submits a bid, b_i, privately to the auctioneer.
  2. The auctioneer decides who gets the good, if anyone.
  3. The auctioneer decides on a selling price.
- Two decision made by the auctioneer: allocation and payment
- Obvious choice for step 2 is the highest bidder. Everyone stated, via their bids, how much they are willing to pay. Allocate the good to the bidder who is willing to pay the most.
- Implementation of Step 3

Example 0 Altruistic. $p = 0$. Bad idea. Since charging nothing turns the auction into who can name the highest number. Leads to an incentive problem.

Example 1 A natural idea. First price auction. Since the auctioneer has asked the bidders the amount they are willing to pay, simply make them pay what they bid. Nontrivial to reason about. Both from the bidder's perspective and the auctioneer's perspective.

Example 2 Right place to start. Second price auction—sealed bid format—highest bidder gets the good and pays the second-highest bid.

- In a second price auction, every bidder has a dominant strategy: set its bid b_i equal to its private valuation v_i. That is, this strategy maximizes the utility of the bidder i, no matter what the other bidders do.
- In second price auction, a truth telling bidder is guaranteed non-negative utility.
- The second price auction simultaneously achieves quite different, but desirable, properties
  1. **Strong incentive guarantees**: It is **dominant strategy incentive compatible (DSIC)**, i.e., the above 2 claims hold
  2. **Strong performance guarantee**: If bidders bid truthfully, then it maximizes social surplus $\sum_i x_i v_i$, where $x_i = 1$ if i wins and 0 if i loses, subject to feasibility constraints $\sum_i x_i \leq 1$.
  3. **Computationally efficient**: The auction can be implemented in polynomial (linear) time.
- **Sponsored Search Auctions**: The goods for sale are the k "slots" for sponsored links on a search results page. The bidders are the advertisers who have a standing bid on the keyword that was searched on. For example, Volvo and Suburu might be bidders on the keyword "station wagon," while Nikon and Canon might be bidders on the keyword "camera."
  Such auctions are more complex than single-item auctions in two ways.
  1. First, there are generally multiple goods for sale (i.e., $k > 1$).
  2. Second, these goods are not identical, slots higher on the search page are more valuable than lower ones, since people generally scan the page from top to bottom.

We quantify the difference between different slots using click-through-rates (CTRs). The CTR α_j of a slot j represents the probability that the end user clicks on this slot. Ordering the slots from top to bottom, we make the reasonable assumption that $\alpha_1 \geq \alpha_2 \geq \ldots \geq \alpha_k$.
For simplicity, we also make the unreasonable assumption that the CTR of a slot is independent of its occupant.
The exercises show how to extend the following results to the more general and realistic model in which each advertiser i has a "quality score" β_i (the higher, the better) and the CTR of the advertiser i in the slot j is the product $\beta_i \alpha_j$.
We assume that an advertiser is not interested in an impression (i.e., being displayed on a page) per se, but rather has a private valuation v_i for each click on its link. Hence, the value derived by the advertiser i from the slot j is $v_i \alpha_j$.

- What's hard about mechanism design problems is that we have to jointly design two things
  1. the choice of who wins what (the allocation)
  2. the choice of who pays what (the payment)
- Interestingly, in many settings, we can tackle this two-prong design problem one step at a time.

Step 1 Assume, without justification, that bidders bid truthfully. Then, how should we assign bidders to slots so that the above properties 2 and 3 hold?

Step 2 Given our answer to Step 1, how should we set selling prices so that the above property 1 holds?

- This two-step approach decouples the allocation rule and the payment rule. In particular, we need to design.
- **Myserson's Lemma** provides a powerful and general tool for implementing Step 2.
  1. Allocation Rule x: Bids $\rightarrow$ Winner. Formally, $x : \mathbb{R}_+^n \mapsto \{0,1\}^n$.
  2. Payment Rule p: Bids $\rightarrow$ Sale price. Formally, $p : \mathbb{R}_+^n \mapsto \mathbb{R}_+$
- Second Price (Vickery) Auction
  1. Bids submitted $b \in \mathbb{R}_+^n$. Write $i^* \in \arg\max_{i \in [n]} b_i$ and $B = \max_{i \neq i^*} b_i$
  2. Highest bidder wins the item and is charges the second-highest bid.
  3. Allocation rule $x_{i^*}(b) = 1$ and $x_i(b) = 0$ for all $i \neq i^*$. Also, $p_{i^*}(b) = B$ and $p_i(b) = 0$ for all $i \neq i^*$.

4.2 Single Parameter Environments

- This framework provides an abstraction to state Myerson's lemma, and captures the three settings (single item auction, multi-item auction, sponsored search auction) we have seen so far.
- Here, we have n agents each with a private valuation $v_i \in \mathbb{R}_+$ which denotes agent $'$s "value per unit stuff" that it gets.
- Finally, in this environment, there is a feasible set $X \subset \mathbb{R}^n$. Each element of X is an n-vector $(x_1, ..., x_n) \in X$, where x_i represents the "amount of stuff" given to agent i.

$$x : \mathbb{R}_+^n \mapsto X \subset \mathbb{R}^n$$

- Quasilinear utility of agent i in single parameter environments is expressed as

$$u_i(b) = v_i x_i(b) - p_i(b)$$

- To ensure individual rationality, we will focus on payment rules which satisfy $p_i(b) \in [0, b_i x_i(b)]$ for every i and bid profile b.
- We first define two properties of allocation rules.
- **Implementable Allocation Rule**: An allocation rule $x : \mathbb{R}_+^n \mapsto X$ of a single-parameter environment is said to be implementable iff there exists a payment rule $p : \mathbb{R}_+^n \mapsto \mathbb{R}_+^n$ such that the sealed-bid auction (x, p) is DSIC.
- The design question can be restated as: Is the allocation rule obtained in step 1 implementable?
- Note that the definition of implementability is "nonoperational," i.e., it is not obvious if, given an allocation rule, we can test whether this property is satisfied or not.

- **Monotone Allocation Rule**: An allocation rule x for a single parameter environment is said to be monotone if for every bidder i and bids b_{-i} of the other agents, the allocation $x_i(x, b_{-i})$ to i is a non-decreasing function of the bid z.

- **Myserson's Lemma**: For a single-parameter environment

  1. An allocation rule x is implementable iff it is monotone.
  2. If an allocation rule x is monotone, then there exists a unique payment rule p such that the auction (x, p) is DSIC (assuming the normalization, $b_i = 0$ implies $p_i(b) = 0$)
  3. The payment rule in 2 is given by an explicit formula.

Consider an allocation rule x, which may or may not be monotone. Suppose there is a payment rule p such that (x, p) is a DSIC mechanism.
Recall the DSIC condition: For every bidder i, every possible private valuation b_i, every set of bids b_{-i} by the other players, it must be that i's utility is maximized by bidding truthfully. For now, fix i and b_{-i} arbitrarily. As shorthand, write $x(z)$ and $p(z)$ for the allocation $x_i(z, b_{-i})$ and payment $p_i(z, b_{-i})$ of i when it bids z, respectively.
We invoke the DSIC constraint via a simple but clever swapping trick. Suppose (x, p) is DSIC, and consider any $0 \leq y < z$. Because bidder i might as well have private valuation z and can submit the false bid y if it wants, DSIC demands that

$$zx(z) - p(z) \geq zx(y) - p(y)$$

Similarly, since bidder i might as well have the private valuation y and could submit the false bid, (x, p) must satisfy

$$yx(y) - p(y) \geq yx(z) - p(z)$$

Myserson's Lemma is, in effect, trying to solve for the payment rule p given the allocation rule x. Rearranging the inequalities yields the following "payment difference sandwich"

$$z[x(y) - x(z)] \leq p(y) - p(z) \leq y[x(y) - x(z)]$$

Thus, we can assume for the rest of the proof that x is monotone.
Fix z and let y tend to z from above. We focus primarily on the case where x is piecewise constant. In this case, x is flat except for a finite number of "jumps". Taking the limit $y \to z$, the left and right-hand sides become 0 if there is no jump in x at z. If there is a jump of magnitude h at z, then the left and right-hand sides both tend to zh. This implies the following constraint on p, for every z

$$\text{jump in } p \text{ at } z = z \cdot \text{ jump in } x \text{ at } z$$

Thus, assuming the normalization $p(0) = 0$, we have derived the following payment formula, for every bidder i, bids b_{-i} by other bidders, and bid b_i by i

$$p_i(b_i, b_{-i}) = \sum_{j=1}^{l} z_j \cdot \text{ jump in } x_i(\cdot, b_{-i}) \text{ at } z_j$$

where $z_1, ..., z_l$ are the breakpoints of the allocation function $x_i(\cdot, b_{-i})$ in the range $[0, b_i]$
A similar argument applies when x is a monotone function that is not necessarily piecewise constant. For instance, suppose that x is differentiable. Dividing the payment sandwich by $y - z$ and taking the limit as $y \to z$ yields the constraint

$$p'(z) = z \cdot x'(z)$$

and, assuming $p(0) = 0$, the payment formula becomes

$$p_i(b_i, b_{-i}) = \int_0^{b_i} z \cdot \frac{d}{dz} x_i(z, b_{-i}) dz$$

for every bidder i, bid b_i, and bids b_{-i} by others.
Thus, for every allocation rule x, there is at most one payment rule p such that (x, p) is DSIC.

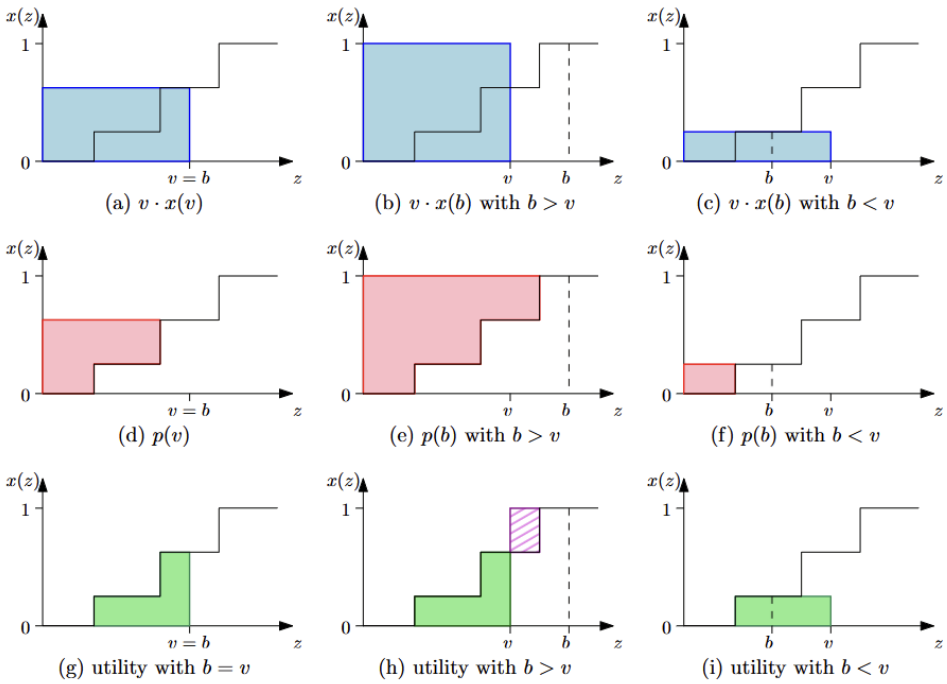


Figure 2: Proof by picture that the payment rule in (6), coupled with the given monotone and piecewise constant allocation rule, yields a DSIC mechanism. The three columns consider the cases of truthful bidding, overbidding, and underbidding, respectively. The three rows show the surplus $v \cdot x(b)$, the payment $p(b)$, and the utility $v \cdot x(b) - p(b)$, respectively. In (h), the solid region represents positive utility and the lined region represents negative utility.

Figure 8: Visual Proof of Myerson's Lemma

- In the sponsored search context the allocation to an agent, let $x(b)$ be the allocation rule that assigns the jth highest bidder to the jth highest slot, for $j = 1, 2, ..., k$. We have noted previously that this rule is surplus-maximizing and monotone. Applying Myerson's Lemma, we can derive a unique payment rule p such that (x, p) is DSIC. Consider a bid profile, b; we can re-index teh bidders so that $b_1 \geq b_2 \geq \ldots \geq b_n$. For intuition, focus on the first bidder and imagine increasing its bid from 0 to b_1, holding the other bids fixed. The allocation $x_i(z, b_{-i})$ ranges from 0 to α_1 as z ranges from 0 to b_1, with a jump of $\alpha_j - \alpha_{j+1}$ at the point where z becomes the jth highest bid in the profile (z, b_{-i}), namely b_{j+1}. Thus, in general, Myerson's payment formula specializes to

$$p_i(b) = \sum_{j=i}^{k} b_{j+1}(\alpha_j - \alpha_{j+1})$$

  for the ith highest bidder, where $\alpha_{k+1} = 0$

- **Revelation Principle**: We roll in two conditions from the definition of DSIC

  1. Every bidder in the auction has a dominant strategy, irrespective of its private valuation.
  2. The dominant strategy is **direct revelation**, wherein the bidder truthfully reports her valuation to the auctioneer.

- **Beyond Dominant-Strategy Equilibria**: There are settings where it is necessary, and also desirable, to consider mechanisms which do not admit dominant-strategy equilibria (e.g., implementation in Bayes-Nash Equilibria). DSIC and non-DSIC mechanisms are incomparable in general, the former enjoy strong incentive guarantees, the latter better performance guarantees.

- The Revelation Principle states that, if we have (designed) a mechanism with Property 1, then we do not need to forgo Property 2.

- For every mechanism M in which every bidder has a dominant strategy (no matter what its private valuation), there is an equivalent direct-revelation DSIC mechanism M'.

- Equivalent in the sense that M and M' have identical an outcome for every valuation profile.

4.3 Revenue Maximization

- Revenue maximization requires a model to reason about trade-offs across different inputs.
- **Bayesian Analysis**: We impose a distribution over the private valuations and seek to do as good as possible in expectation with respect to those distributions. That is, we cannot obtain a single auction which maximizes revenue over the set of all inputs. But, we do have auctions that maximize expected revenue with respect to specified distributions.
- **The Bayesian Model** consists of
  1. Single-parameter environment
  2. The valuation v_i is independently drawn from a distribution F_i (with density function f_i) with support $[0, v_{max}]$
  3. The distributions are known to the mechanism designer. Though, as usual, the realization v_i is known only to bidder i.
- In this Bayesian environment the "optimal" auction is clearly defined: it is the auction, among all DSIC auctions, which maximizes expected revenue.
- The expectation is with respect to the given distribution $F_1 \times F_2 \times ... \times F_n$ over the valuation profiles (assuming truthful bidding).
- The expected revenue of a posted price r is simple $r(1 - F(r))$. Given a distribution F, finding the optimal value of r leads to the optimal auction.
- The optimal posted prices is called the **monopoly price** of the distribution F.
- **Single Item Two Bidders**
  In the Bayesian Setup, consider the bidders with $F_1 = F_2 = Unif[0, 1]$
  The second price auction has nonzero expected revenue. Specifically, revenue of second price auction $= \sum_{i=1}^{2} p_i(b)$
  Assuming, truth telling, revenue = second-highest valuation.
  Expected Revenue $= E_{X,Y \sim Unif(0,1)}[\min\{X, Y\}] = \frac{1}{3}$
- Another option is to supplement the second-price auction with a **reserve price**, r.
  In this auction, if the bids are both less than r, then no one gets the good. Else, the highest bidder gets the good at price $\max\{r, \text{second highest price}\}$
  This is equivalent to an auction where we introduce a fictitious bid of r and charge it zero in case r wins
- Note that the expected value of n iid uniform rv, $X_1, X_2, ..., X_n \sim_{iid} Unif(a, b)$, $Y = \min_i X_i$, then $E[Y] = \frac{b+na}{n+1}$
- **Goal**: Description of a revenue maximizing auction for any single parameter environment and any distribution(independent) $F_1, F_2, ..., F_n$
- Throughout, we focus on DSIC mechanisms. Hence, assume truthful bids, $b_i = v_i \ \forall i$
- By definition, the expected revenue is

$$E_{(v_i \sim F_i)_i}[\sum_{i=1}^{n} p_i(v_1, v_2, ..., v_n)]$$

- We will develop a formula for expected revenue that only references the allocation rule x and hence is far easier to maximize.
- **Virtual Valuation**: For bidder i with value distribution F_i and valuation v_i, their virtual valuation $\varphi_i(v_i)$ is defined as

$$\varphi_i(v_i) = v_i - \frac{1 - F_i(v_i)}{f_i(v_i)}$$

  Note $\varphi_i(z)$ can be negative
- $\varphi_i(z) \leq z$ and $\frac{1-F_i(z)}{f_i(z)}$ is sometimes called the "information lost".
- Expected Revenue equals Expected Virtual Welfare

- For every single parameters with valuation distributions $F_1, F_2, ..., F_n$, every DSIC mechanism. (x, p), every i, every value v_{-i} of the other agents

$$E_{v_i \sim F_i}[p_i(v)] = E_{v_i \sim F_i}[\varphi_i(v_i)x_i(v)]$$

  Note $v = (v_i, v_{-i})$

- We start with Myerson's Lemma

$$\begin{aligned}
p_i(v) &= \int_0^{v_i} z x_i'(z, v_{-i}) dz \\
\text{Step 1: } E_{v_i \sim F_i}[p_i(v)] &= \int_0^{v_{max}} p_i(v) f_i(v_i) dv_i \\
&= \int_0^{v_{max}} \left[\int_0^{v_i} z x_i'(z, v_{-i}) dz \right] f_i(v_i) dv_i
\end{aligned}$$

  Step 2: Reverse the order of integration

$$= \int_0^{v_{max}} \left[\int_z^{v_{max}} f_i(v_i) dv_i \right] z x_i'(z, v_{-i}) dz$$

  This simplifies to

$$= \int_0^{v_{max}} [1 - F_i(z)] z x_i'(z, v_{-i}) dz$$

  $[1 - F_i(z)]z = g$ and $x_i'(z, v_{-i}) = h'$

  Step 3: Integration by parts yields

$$= (1 - F_i(z)) z \cdot x_i(z, v_{-i}) \Big|_0^{v_{max}} - \int_0^{v_{max}} x_i(z, v_{-i})(1 - F_i(z) - z f_i(z)) dz$$

  Hence, we get

$$\begin{aligned}
&= \int_0^{v_{max}} \left(z - \frac{1 - F_i(z)}{f_i(z)} \right) x_i(z, v_{-i}) f_i(z) dz \\
&= \int_0^{v_{max}} \varphi_i(z) x_i(z, v_{-i}) f_i(z) dz
\end{aligned}$$

  This quantity corresponds to $E_{v_i \sim F_i}[\varphi_i(v_i)x_i(v_i, v_{-i})]$
  Therefore, $E_{v_i \sim F_i}[p_i(v)] = E_{v_i \sim F_i}[\varphi_i(v_i)x_i(v_i, v_{-i})]$

- For every single parameter environment with valuation distributions $F_1, F_2, ..., F_n$ and every DSIC mech (x, p)

$$E_{v \sim F}[\sum_{i=1}^n p_i(v)] = E_{v \sim F}[\sum_{i=1}^n \varphi_i(v_i)x_i(v)]$$

- Taking the expectation w.r.t $v_i \sim F_{-i}$ in the previous Lemma's equality

$$E_{v \sim F}[p_i(v)] = E_{v \sim F}[\varphi_i(v_i)x_i(v)]$$

  Linearity of expectation gives us

$$\begin{aligned}
E_{v \sim F}[\sum_{i=1}^n p_i(v)] &= \sum_{i=1}^n E_{v \sim F}[\varphi_i(v)] \\
&= \sum_{i=1}^n E_{v \sim F}[\varphi_i(v_i)x_i(v)] \\
&= E_{v \sim F}[\sum_{i=1}^n \varphi_i(v_i)x_i(v)]
\end{aligned}$$

- **Welfare Maximizing Allocation**: $(\hat{x}, \hat{p})$
  1. Take in bids $v_1, v_2, ..., v_n$
  2. Find

$$\underset{(x_1, x_2, ..., x_n) \in X}{\arg\max} \sum_{i=1}^n x_i v_i$$

  3. Change payment $\hat{p}$ according to Myerson'e Payment Formula.

- **Virtual Welfare Maximizing Allocation**: $(\tilde{x}, \tilde{p})$
  1. Take in bids $(v_1, v_2, ..., v_n)$
  2. Compute $\varphi_i(v_i)$
  3. Find
  $$\underset{(x_1,x_2,...,x_n)\in X}{\arg\max} \sum_{i=1}^{n} x_i \varphi_i(v_i)$$
  4. Change Payment $\tilde{p}$ according to Myerson's Payment Formula.
- A distribution with CDF F_i is said to be regular if the corresponding virtual value function $\varphi_i(\cdot)$ is non-decreasing.
- For regular distributions $F_1, F_2, ..., F_n$, the virtual welfare maximization auction is DSIC and revenue optimal.
- Consider a single item auction with identical value distributions $(F_1 = F_2 = ... = F_n)$
  Revenue optimal auction = Second Price auction with reserve price $r = \varphi^{-1}(0)$
- However, even for single item setting with non-identical distributions the auction is relatively complex, requires detailed info about the value distribution, and does not resemble auctions in practice.
- Pursue **approximately optimal** auctions that are simpler, more practical and more robust.
- The complexity is inevitable if one waits to maximize expected revenue up to the last bit.
- **Optimal Stopping Rules**
  1. Known Distributions(Independent): $G_1, G_2, ..., G_n$ and number of rounds n
  2. In round $i \in \{1, 2, ..., n\}$, observe realization $\pi_i \sim G_i$
  3. After seeing π_i, irreversible decision
     Accept π_i, Stop
     Discard π_i(permanently), continue
- Trade off under uncertainty: Risk of accepting a reasonable prize π_i early and then missing out later vs Risk of having to settle for a lower profit in end stages.
- Prophet Inequality offers a simple strategy that performs as well as a fully clairvoyant 'prophet'
  $$\text{Benchmark } E_{\pi_i \sim G_i}[\max_{1\le i\le n} \pi_i]$$
- **Prophet's Inequality**: For every sequence of independent distributions $G_1, G_2, ..., G_n$, there is a strategy that guarantees expected reward at least $\frac{1}{2}E_{\pi\sim G}[\max_i \pi_i]$. Moreover, there is such a threshold strategy, which accepts prize i iff π_i is some threshold t.
  Write $z^+ = \max\{0, z\}$
  Consider any threshold t. We derive an upper bound on E max and a lower bound on the expected reward of threshold t.
  $q(t)$ = Probability that all π_i's are less than t, i.e., the threshold strategy accepts no reward.
  Reward of threshold t strategy is 0 w.p. $q(t)$ and $\ge t$ w.p. $1 - q(t)$, we refine the second case.
  Event E_i: Exactly prize $\pi_i \ge t$, then, beyond t, the reward additionally has $(\pi_i - t)$
  When two rewards π_i and π_j are more than t, then the reward gained depends on which of π_i/π_j is observed first. In this analysis, we avoid this complication and only go beyond t, when exactly one of the rewards are greater than t.

$$\begin{aligned}
E_{\pi\sim G}[\text{Payoff of } t\text{-threshold strategy}] &\ge (1-q(t))t + \sum_{i=1}^{n} E_\pi[\pi_i - t|E_i]Pr\{E_i\} \\
&= (1-q(t))t + \sum_{i=1}^{n} E_\pi\left[\pi_i - t \middle| \begin{array}{l} \pi_i \ge t \\ \pi_j < t\ \forall j \ne i \end{array}\right] Pr\left\{ \begin{array}{l} \pi_i \ge t \\ \pi_j < t\ \forall j \ne i \end{array} \right\} \\
&= (1-q(t))t + \sum_{i=1}^{n} E_\pi\left[\pi - t \middle| \begin{array}{l} \pi_i \ge t \\ \pi_j < t\ \forall j \ne i \end{array}\right] Pr\{\pi_i \ge t\}Pr\{\pi_j < t\ \forall j \ne i\} \\
&= (1-q(t))t + \sum_{i=1^n} E_\pi[\pi_i - t|\pi_i \ge t]Pr\{\pi_i \ge t\}Pr\{\pi_j < t\ \forall j \ne i\} \\
&\ge (1-q(t))t + q(t)\sum_{i=1}^{n} E[(\pi_i - t)^+]
\end{aligned}$$

Therefore,

$$E_{\pi\sim G}[\text{Payoff of } t\text{-threshold strategy}] \geq (1-q(t))t + q(t)\sum_{i=1}^{n} E[(\pi_i - t)^+]$$

Upper Bound on Prophet's Expected Reward

$$\begin{aligned} E_\pi[\max_i \pi_i] &= E_\pi[t + \max_{1\leq i\leq n}(\pi_i - t)] \\ &\leq E_\pi[t + \max_{1\leq i\leq n}(\pi_i - t)^+] \\ &\leq t + E_\pi[\sum_{i=1}^{n}(\pi_i - t)^+] \\ &= t + \sum_{i=1}^{n} E_\pi[(\pi_i - t)^+] \end{aligned}$$

Comparing the LB and UB, we can set t such that $q(t) = \frac{1}{2}$ to obtain the threshold that satisfies the theorem statement.
Works with Adversarial tie breaking.

- $\pi_i = \varphi_i^+(v_i)$, φ_i is the corresponding distribution induces by F_i.
  In the current context of single item auction, with regular F_i's.

$$\begin{aligned} \text{Expected revenue of optimal auction } &= E_{v\sim F}[\sum_{i=1}^{n}\varphi_i(v_i)x_i(v)] \\ &= E_{v\sim F}[\max_i \varphi_i^+(v_i)x_i(v)] \end{aligned}$$

  Hence, the expected optimal revenue = expected prize obtained by a prophet.

- **Virtual Threshold Allocation Rule**
  1. Choose t such that $Pr[\max_i \varphi_i^+(v_i) \geq t] = \frac{1}{2}$
  2. Give the item to bidder i with $\varphi_i(v_i) \geq t$, if any, breaking ties arbitrarily.

- Virtual threshold allocation rules are near optimal
  If x is virtual threshold allocation rule, then

$$E_v[\sum_{i=1}^{n}\varphi_i^+(v_i)x_i(v)] \geq \frac{1}{2}E_v[\max_i \varphi_i^+(v_i)]$$

- **Second-Price with Bidder-Specific Reserves**
  1. Set a reserve price $r_i = \varphi_i^{-1}(t)$ for each bidder i with t defined as the virtual threshold allocation rules.
  2. Give the item to the highest bidder that meets their reserve, if any.

  Under regular distribution, this is a monotone allocation rule.

- **Simple vs Optimal Auction**: For all $n \geq 1$ and regular distributions $F_1, F_2, ..., F_n$, the expected revenue of a second price auction, with suitable reserve prices is at least $\frac{1}{2}$ of that of the optimal auction.

- A critique of optimal mechanism for revenue maximization: The value distributions $F_1, F_2, ..., F_n$ are assumed to be known to the mechanism designer in advance.

- **Prior-Independent Mechs**: Still conferring to a Bayesian Analysis. We continue to assume that the valuations are draw from distributions. It is just that the distributions are unknown to the designer. Distribution is used in the analysis, but not in the design.

- **Bulow-Klemperer Theorem**: The expected revenue of an optimal single item auction is at most that of a second-price auction (no reserve price) with one extra bidder.
  Let F be a regular distribution and n a positive integer. Let p and p^* be the payment vectors of the second price auction with $(n+1)$ bidders and the optimal auction with n bidders, respectively. Then,

$$E_{v\sim F^{n+1}}[\sum_{i=1}^{n} p_i(v)] \geq E_{v\sim F^n}[\sum_{i=1}^{n} p_i^*(v)]$$

Informally,

$$E_{v\sim F^{n+1}}[\text{Rev-Vickery}(n+1)] \geq E_{v\sim F^{n}}[\text{Rev-Virtual Welfare Maximization}(n)]$$

This gives us a prior independent guarantee. Also, second-price auction simultaneously works for all regular F.
Comparison via a fictitious auction F
$A \leftarrow (n+1)$ DSIC auction

1. Simulate an optimal n-bidder auction for F on the first n bidders. $1, 2, ..., n$
2. If the item is not awarded in the first step, then give the item to the bidder $(n+1)$ for free.

Two properties of A

1. Expected revenue of A equals that of an optimal auction with n bidders
2. It always allocates the item.

We note that expected revenue of a second-price auction(with $n+1$ bidders) is at least that of A.
All allocations that allocate the item, $\max_{1\leq i\leq n} \varphi(v_i)$. Since, φ is regular and identical, this becomes $\max_{1\leq i\leq n} v_i$ which is exactly the second-price auction.
Hence, Expected Revenue of Second-Price$(n+1) \geq$ Expected Revenue $A(n+1) \geq$ Expected Revenue Optimal(n)

- So far, private valuation of each agent i has been a single number v.
- **Multi-Parameter Mechanism Design**: Here, Cardinal preference of each agent is much more expressive.
- For multi-parameter environments, the **Vickrey-Clarke-Groves(VCG) Mechanism** provides a sweeping positive result: DSIC welfare maximization is possible(in principle) in every multi-parameter environment.
- **General Mechanism Design Environments**: Components

  1. n bidders
  2. A finite set Ω of 'outcomes'
  3. Each bidder $i \in (n)$ has a private valuation $v_i(\omega)$ for each outcome $\omega \in \Omega$

  The set Ω is abstract and can be very large.
  The social welfare of an outcome $\omega \in \Omega$ is defined as

$$\sum_{i=1}^{n} v_i(\omega)$$

- Consider, the single item auction, here $|\Omega| = n+1$

$$\Omega = \{\omega_0, \omega_1, ..., \omega_n\}$$

  ω_0: No agent receives the item
  ω_i: Agent i receives the item
  Here, $v_i(\omega_i) > 0$ and $v_i(\omega_j) = 0\ \forall j \neq i$

- **Combinatorial auction**: A set of m items(indivisible) and bidders can have complex preferences between different subsets of items.
  $\Omega \leftarrow$Set of all n partitions of m
  $(\delta_1, \delta_2, ..., \delta_n) \in \Omega$, denoting that subset $\delta_i \subseteq [m]$ assigned to bidder i.
  $v_i(s) \leftarrow$value of bidder i for subset (m)
  Here, each bidder has 2^m private valuations.
- Every multi-parameter environment admits a DSIC welfare-maximizing allocation
- In every general mechanism design environment, there is a DSIC welfare-maximizing mechanism.
- We follow the two-step approach that worked in the single parameter environment
- Assume, without justification, that bidders report their private valuation, truthfully. Then, figure out which outcome to pick.
  In welfare maximization; using bids as proxies for valuations(unknown).
  Define allocation rule x by

$$x(b) = \arg\max_{\omega\in\Omega} \sum_{i=1}^{n} b_i(\omega)$$

  b_i, like v_i, is a vector indexed by Ω

- The second step is to define the payment rule p, which when coupled with x, yields a DSIC mechanism
  Key Idea: Set payment for i as the "externality" caused by i.
  The welfare loss inflicted on the other $(n-1)$ bidders by i's presence.
  Specifically,
  $$p_i(b) = (\max_{\omega \in \Omega} \sum_{j \neq i} b_j(\omega)) - \sum_{j \neq i} b_j(\omega^*)$$
  where $\omega^* = x(b)$ is chosen previously.
  Aligning the incentive of i with a welfare maximizing decision, $p_i(b) \geq 0$.

- **VCG Mechanism**: A mechanism with allocation and payment rules (x, p) as defined previously.
  Alternate form of payment
  $$p_i(b) = b_i(\omega^*) - \left[\sum_j b_j(w^*) - \max_{\omega \in \Omega} \sum_{j \neq i} b_j(\omega) \right]$$
  rebate, increase in social welfare attributable to i's presence.
  $p_i(b) \leq \frac{b_i}{\omega^*}$, hence, individually rational.

- By definition VCG mechanism (x, p) maximizes social welfare when the bidders are truthful. We complete the proof by establishing DSIC.
  That is, for every bidder i, every set of reports b_{-i} by the other bidders, the agent i maximized the quasilinear utility $v_i(x_i(b)) - p_i(b)$ by setting $b_i = v_i$.
  Fix i and b_{-i}, when the chosen outcome $x(b)$ is ω^*, agent i's utility is
  $$v_i(\omega^*) - p_i(b) = [v_i(\omega^*) + \sum_{j \neq i} b_j(\omega^*)] - [\max_{\omega \in \Omega} \sum_{j \neq i} b_j(\omega)]$$
  First term is A and second term is B.
  The term B is a constant, independent of i's report b_i.
  Hence, the problem of maximizing agent i's utility reduces to maximizing A.
  $$b_i = v_i \xrightarrow{x} \omega^*$$
  $$b_i' \xrightarrow{x} \omega'$$
  Thus, truthful reporting aligns i's objective with the mechanism's objective.

- If I don't bid my valuation, the VCG mechanism rule ensures that you don't get max utility.

- **Robertson's Theorem**: Uniqueness of payment rule for multiparameter environment

5 Contract Theory

- Economic tool to incentivize a strategic agent to put in effort.

- Deals with "How to incentivize people to put in effort"

- **Basic Model** [Holmstrom '79]

  1. Two players: Principal and agent
  2. Ingredients: Private Information and Incentives

  Need a right level of compensation to get a balance between how much the prinicpal has to pay and the agent's effort.

- Agent's actions are hidden - "Moral Hazard"

- Principal never charges, only pays the agent - "Limited Liability"

- **Moral Hazard**: Situation in which the agent is executing some action, but not reaping the reward of the action. The principal is getting the reward.

- **Design Goal**: Maximize objective(welfare/revenue) subjective to incentive constraints. Truthtelling or effort.

- **Contract Design under Uncertainity**: Model

  1. Agent has n actions(effort levels) with costs $C_1, C_2, ..., C_n$

2. Principal has m rewards $r_1, r_2, \ldots, r_m \in \mathbb{R}_+$
3. Every action i induces distribution F_i over r
   F_{ij} = Probability that action a_i yields reward r_j
   $R_i = \sum_j F_{ij} r_j$ = Expected reward from action i

$$\begin{bmatrix} & r_1 & r_2 & \cdots & r_m \\ 1 & & —F_1— & & \\ 2 & & —F_2— & & \\ \vdots & & \vdots & & \\ n & & —F_n— & & \end{bmatrix}$$

- Intuitively, higher cost should result in more F_{ij} for higher rewards
- Can be tought of as the tuple $(C, r, (F_1, \ldots, F_n))$
- **Contract**: Vector of transfers t, specifies a payment $t_j \geq 0$ per reward r_j

$$T_i = \sum_j t_j F_{ij} = \text{ Expected payment for action } i$$

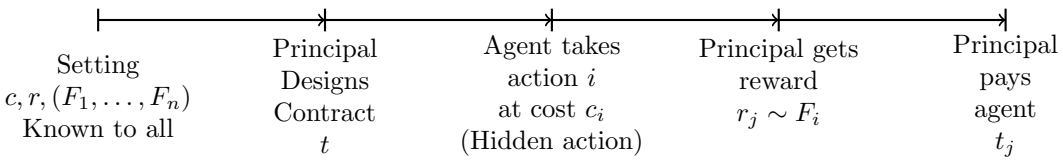
- Transfer depends on reward. NOT on effort level.
- Agent selects effort i that maximizes their expected utility

$$T_i - C_i = \text{Expected Payment} - \text{Cost}$$

- Principals's expected revenue for agent choice $i \in [n]$

$$R_i - T_i = \text{Expected Reward} - \text{Expected Payment}$$

- Timeline



- We say that a contract is **optimal** if it maximizes principal's expected revenue.
- **Minimum Payment Problem**
  Input: Contract setting (C, r, F), and action i
  Output: Minimum T_i that incentivizes action i

$$\begin{aligned} \text{LP:} \quad & \min_{(t_1, t_2, \ldots, t_m)} \sum_j F_{ij} t_j \\ \text{st.} \quad & \sum_j F_{ij} t_j - C_i \geq \sum_j F_{aj} t_j - C_a \quad \forall a \in [n] \end{aligned}$$

  Optimal contract is solvable via n minimum payment problems

- **Implementability Problem**: Given action i, does there exist a contract t such that under it, action i is utility maximizing for the agent.
- **Dual of Minimum Payment**
  Lemma: Action i is implementable (upto tie breaking) iff no convex combination of the other actions has same distribution over the rewards at a lower cost.
  Primal feasible: Action i is implementable iff following LP is feadible

$$\begin{aligned} & \min 0 \\ \text{such that } & \sum_j F_{ij} t_j - C_i \geq \sum_j F_{aj} t_j - C_a \quad \forall a \in [n], a \neq i \end{aligned}$$

Primal Infeasible iff $\exists$ feasible dual solution with objective > 0

$$\begin{aligned} &\max C_i - \sum_{a \neq i} \lambda_a C_a \\ \text{such that } &\sum_{a \neq i} \lambda_a F_{aj} \leq F_{ij} \quad \forall j \in [m] \\ &\lambda_a \geq 0 \quad \sum_{a \neq i} \lambda_a = 1 \end{aligned}$$

- There exists an optimal contract with $\leq (n-1)$ non-zero transfers
- **Criticisim of LP-Based Approach**
  1. Requires perfect knowledge of F, "More normative than Positive"
  2. Dependence on n and m might be prohibitive
  3. The contract that comes out of LP might be arbitrary
- **Optimal contracts for 2 actions and 2 rewards**: $n = m = 2$

		Failure r_1	$<$	Success r_2
(low)	$C_1 = 0$	$1-p$		p
(High)	$C_2 = C > 0$	$1-\pi$		π

Principal can always ectract $R_1 = (1-p)r_1 + pr_2$
First-Best= $R_2 - C = r_1(1-\pi) + r_2\pi - C =$ Assume that optimal contract incentivizes action 2
Contract pays t_1 for failure and t_2 for success

$$\begin{aligned} t_1(1-\pi) + t_2\pi - C &\geq t_1(1-p) + t_2 p \\ (\pi - p)(t_2 - t_1) &\geq C \end{aligned}$$

To minimize transfer, must have $t_1 = 0$, which gives us $t_2 = \frac{C}{\pi - p}$
Hence, expected revenue of principal $= R_2 - \frac{C}{\pi-p}\pi$
Optimal contract has a "closed form" here
Monotonicity holds here, $t_2 > t_1$, transfer increases ith reward
Generalizes to any n as long as $m = 2$

- **Optimal conract 2 actions, m rewards**

	r_1	r_2	$\cdots$	r_m
Action 1	F_{11}	F_{12}	$\cdots$	F_{1m}
Action 2	F_{21}	F_{22}	$\cdots$	F_{2m}

Cost for each action is still 0 and C
Binding IC constraint for action 2, for some j

$$t_j F_{2j} - C = t_j F_{1j}$$

Optimal contract is $t_j = \frac{C}{F_{2j} - F_{1j}}$
Principla extracts $= R_2 - \frac{C}{F_{2j} - F_{1j}} F_{2j}$, that is

$$R_2 - C \frac{1}{\left(1 - \frac{F_{1j}}{F_{2j}}\right)}$$

Hence, choose j^* that minimizes $\frac{F_{1j}}{F_{2j}}$
This ratio is called the likelihood ratio between actions 1 and 2
Optimal contract pays for reward with minimum likelihood ratio.

- Role of reward in the model is two-fold
  1. Represent surplust to be shared
  2. Signal to the principal the agents action

The optimal contract is shaped by (2), can be mismatched with (1)